# Exchange Rates under Fiscal Dominance

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**Abstract**

We propose a framework to study exchange rates and their relationship with macroeconomic aggregates under alternative monetary–fiscal policy interactions. We first develop a simple flexible-price model and derive closed-form solutions for nominal and real variables. When fiscal policy is active and future surpluses do not stabilize the real value of public debt, fiscal shocks are unfunded and lead to a depreciation of the nominal exchange rate without affecting real variables. We then develop a quantitative model with a cash-in-advance constraint, through which unfunded fiscal shocks also affect real variables, including the real exchange rate. When taken to the data, our framework replicates the dynamics of nominal and real macroeconomic aggregates in Brazil over 1980–2019, a period marked by transitions between monetary and fiscal dominance.

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# 1 Introduction

*Persistent high inflation is always and everywhere a fiscal phenomenon, in which the central bank is a monetary accomplice. [...] Persistent inflation is caused by a coordinated monetary and fiscal policy regime that calls for the central bank persistently to print currency to supply the fiscal authority with revenues via an implicit "inflation tax."*

Thomas Sargent, "Letter to Another Brazilian Finance Minister,"
*Banco de Investimentos Garantia S.A. Newsletter* (November, 1993)

The level of public debt in advanced economies and the growing doubts about its sustainability have put monetary-fiscal policy coordination back at the center of the inflation debate. Under fiscal dominance, when the primary balance is set without regard to the outstanding debt, the central bank is left with no real choice about inflation: the deficit has to be financed, and part of the financing must come from the increase in the price level. Since the seminal work of Sargent and Wallace (1981), Leeper (1991), and Woodford (1995), what this implies for inflation has been studied at length. Instead, what it implies for nominal and real exchange rates has received much less attention, and this is the question we address in this paper.

We develop a theoretical framework that integrates insights on the role of international financial flows in exchange-rate determination from international macro-finance (Gabaix and Maggiori 2015) with those recently provided by the closed-economy monetary model of Bianchi, Faccini, and Melosi (2023) who distinguish between funded fiscal shocks, which future surpluses will cover, and unfunded ones, which must be absorbed by changes in the price level. Bianchi, Faccini, and Melosi (2023) show that unfunded fiscal shocks account for much of US inflation. We take their distinction into an open-economy model and ask what it implies for exchange rates.

We find that an unfunded increase in the deficit raises inflation and depreciates the nominal exchange rate, whereas the real exchange rate hardly moves. The two forces the international macro-finance literature has emphasized do not behave this way: productivity disturbances and financial shocks to the terms on which a country borrows and lends abroad tend to move the nominal and real exchange rates in the same direction. Among the disturbances we consider, the unfunded fiscal shock is the only one that generates a wedge between the two.

We develop our analysis in two steps. We first propose a simple cashless model under flexible prices to derive a closed-form solution that transparently links the nominal exchange rate to unfunded fiscal shocks, as well as to productivity disturbances and international financial flows. In this model, international financial markets are complete and the Backus and Smith (1993) risk-sharing condition holds up to a shock, which we label an international financial shock. This shock plays the same role as the currency demand shock in Gabaix and Maggiori (2015) and is absent under a pegged exchange rate: in the absence of exchange-rate risk, the UIP condition holds exactly, leaving no room for a currency demand shock.

The simple model is useful to illustrate the main mechanisms at work in our framework, but does not allow unfunded fiscal shocks to also affect real variables, including the real exchange rate. We then develop a quantitative model with incomplete financial markets à la Gabaix and Maggiori (2015) and a cash-in-advance constraint that breaks the dichotomy between nominal and real variables, allowing unfunded fiscal shocks to also affect real variables, including the real exchange rate.

We consider three regimes that arise from alternative monetary–fiscal policy interactions. In the first regime (which for simplicity we label '*floating*'), the central bank follows a Taylor rule targeting inflation, while the government follows a fiscal rule that stabilizes public debt, with fiscal shocks being funded. The second regime (which for simplicity we label '*pegged*') differs from the first in that the central bank implements a currency peg. In both regimes, monetary policy is active and fiscal policy is passive, in the sense of Leeper (1991): the government adjusts future surpluses to stabilize public debt, regardless of the path of inflation, leaving monetary policy free to pursue its own target. In the third regime (which for simplicity we label '*fiscal*'), the Taylor principle is not satisfied, and the primary deficit is subject to unfunded fiscal shocks. In this regime, we have fiscal dominance: fiscal policy is active, while monetary policy is passive.

Under the *floating* regime, the nominal and the real exchange rates move together, driven by volatile international financial shocks; under the *pegged* regime, international financial shocks are absent, and the volatility of both exchange rates declines, with productivity shocks being the only remaining source of variation. Instead, under fiscal dominance, the fiscal shock is not funded, and hence raises inflation to satisfy the government budget constraint. Since the real exchange rate, being a real variable, does not respond, this translates into a depreciation of the nominal exchange rate. If fiscal shocks are sufficiently volatile, they can weaken the tight comovement between the nominal and real exchange rates. Our closed-form expression for the nominal exchange rate, quite surprisingly, shows that its elasticity to the fiscal shock depends negatively on the level of public debt: a smaller debt base requires a larger rise in inflation and, hence a larger depreciation of the nominal exchange rate to absorb a given fiscal shock.

We bring this mechanism to the data through the lens of Brazil's historical experience, as described in Kehoe and Nicolini (2022). Brazil provides a natural laboratory, having experienced all three regimes in sequence: a period of fiscal dominance characterized by high inflation and a floating exchange rate from the 1980s until mid-1994; a pegged exchange rate against the U.S. dollar under the fiscal stabilization program of the *Plano Real* until the end of 1998; and a floating exchange rate thereafter. Comparing the fiscal-dominance regime with the monetary-dominance regime that has prevailed under floating exchange rates since 1999, we find that annual inflation averaged 353% under the former and 6.3% under the latter. At the same time, whereas the nominal and real exchange rates have moved almost one-for-one under the monetary-dominance regime, with a correlation of 1.00, their correlation was only 0.55 under fiscal dominance. By contrast, under fiscal dominance, the nominal exchange rate was more closely correlated with inflation, with a correlation of 0.67, compared with 0.27 after 1999.

We first calibrate the simple model with Brazilian data. The model reproduces the weakening in the comovement between nominal and real exchange rates: the correlation between nominal and real depreciation is 0.62 in the *fiscal* regime, compared with 0.55 in the data, and 1.00 in the *floating* regime, matching the data exactly. It also reproduces the relative volatilities of inflation, output, and exchange rates. The variance decomposition echoes the findings of Bianchi, Faccini, and Melosi (2023) and extends them to an open-economy framework. In the fiscal regime, unfunded deficit shocks account for 89% of the variance of inflation and 56% of the variance of the nominal exchange rate. Under the *floating* regime, they account for none of the variance, while international financial shocks account for 89% of the variance of both the nominal and real exchange rates. These results carry over to the quantitative model, which features two countries, capital, imported intermediates, and strategic complementarities in price setting, in addition to incomplete financial markets à la Gabaix and Maggiori (2015) and a cash-in-advance constraint that allows unfunded fiscal shocks to affect real variables.

From our analysis, we draw two implications. The first is that whether nominal and real exchange rates move together depends on the monetary-fiscal policy interaction, and hence on which shocks dominate, not only on the exchange-rate arrangement. The classic Mussa puzzle is also present in the Brazilian data: the volatility of the real exchange rate rises and falls with that of the nominal exchange rate around the two changes in the exchange-rate regime, in 1994 and 1999. What our framework adds is that the monetary-fiscal policy interaction can give rise to a regime of fiscal dominance, in which unfunded fiscal shocks drive the nominal exchange rate. The second implication is that the *Plano Real* reduced exposure to international financial shocks through the adoption of a currency peg and implemented a fiscal consolidation that reduced the role of unfunded fiscal shocks. Our framework allows us to distinguish between these two channels and suggests that, for the nominal exchange rate, the fiscal channel may be at least as important as the international financial channel.

**Related Literature.** As mentioned above, our framework builds on two influential contributions: Gabaix and Maggiori (2015) in international macro-finance and Bianchi, Faccini, and Melosi (2023) in monetary economics. From the former, we adopt the concept of segmentation in financial markets, which generates deviations from the uncovered interest parity (UIP). From the latter, we adopt the notion of unfunded fiscal shocks, which they show have historically been a key determinant of US inflation. In doing so, our paper contributes to three strands of literature.

First, our work connects with the strand of literature on monetary-fiscal policy interactions in closed-economy models, whose origins can be traced to Sargent and Wallace (1981) and which was subsequently developed by Leeper (1991) and Woodford (1995) (see Canzoneri, Cumby, and Diba, 2010 for a survey of the first wave of this strand). More recently, this line of research has been revitalized by a series of closed-economy studies by John Cochrane (Cochrane 2022, 2024), together with his textbook treatment on the fiscal theory of the price level (Cochrane, 2023). The recent contribution of Bocola et al. (2026) is particularly relevant to our analysis, as it develops a closed-economy model of endogenous fiscal and monetary policy using the same data source as ours (Kehoe and Nicolini 2022).

Second, our paper contributes to the strand of literature analyzing monetary-fiscal policy interactions in open-economy models by showing that fiscal shocks can be a first-order driver of exchange rate dynamics under fiscal dominance. An earlier part of these studies integrates the fiscal theory of the price level into models of currency crises, motivated in part by the Asian financial crisis of the late 1990s (Burnside, Eichenbaum, and Rebelo, 2001; Daniel, 2001*a*) and the European sovereign debt crisis (Daniel and Shiamptanis, 2012). Other papers incorporate the fiscal theory of the price level into models of monetary unions (Bergin, 2000; Canzoneri, Cumby, and Diba, 2001) or focus on the determination of exchange rates under the fiscal theory of the price level without analyzing unfunded fiscal shocks (Loyo, 1999, Daniel, 2001*b*, and Dupor, 2000). More recently, Bianchi (2021) and Alberola et al. (2022) have studied the possibility of default in an open-economy model under fiscal dominance, while Witheridge (2024) and Iorio and Garcia-Cicco (2026) have simulated the effects of monetary and fiscal shocks in open-economy models with sticky prices. A work close to ours is Jiang (2022), which likewise derives a closed-form solution linking the real exchange rate to fiscal shocks through the intertemporal budget constraint of the government. However, the mechanism differs from ours: fiscal shocks are absorbed by real discount rates and the *real* exchange rate, which comoves with the nominal exchange rate due to the nominal rigidities assumed by Jiang (2022). By contrast, we assume flexible prices, so fiscal shocks are absorbed by inflation and fall entirely on the *nominal* exchange rate, for which we provide a closed-form solution, thereby breaking its comovement with the real exchange rate that previous works have attributed to productivity and international financial shocks.

Third, our paper connects to the strand of literature on exchange rates and macroeconomic aggregates. The exchange rate disconnect documented by Meese and Rogoff (1983) and the regime-dependent volatility of the real exchange rate documented by Mussa (1986) have long challenged open-economy models (Baxter and Stockman, 1989; Cole and Obstfeld, 1991; Backus and Smith, 1993; Obstfeld and Rogoff, 2000). One line of research studies the role of different exchange-rate regimes (Monacelli, 2004; Benigno and Benigno, 2008; Flaccadoro and Nispi Landi, 2022; Itskhoki and Mukhin, 2025). Another focuses on the determinants of exchange rate fluctuations in the presence of deviations from the UIP (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021; Lilley et al., 2022; Kekre and Lenel, 2024), as well as on whether the exchange-rate regime is sufficient to preserve monetary autonomy in the presence of a global financial cycle (Rey, 2015; Farhi and Werning, 2014; Banerjee, Devereux, and Lombardo, 2016). We contribute to this strand of literature by developing a unified model that nests three empirically relevant regimes—*floating*, *peg*, and *fiscal*—allowing us to compare them within a common framework and isolate the role of the fiscal stance.

# 2 Simple model

We develop a simple model by augmenting a real business cycle model to incorporate international financial shocks. The home country is modeled as a small open economy, while the rest of the world (henceforth, the foreign economy, denoted by a superscript'*') is treated as exogenous. A representative household maximizes utility by choosing consumption, labor supply, and bond holdings. Under flexible prices, a representative firm produces the domestic good using labor and exports it to the foreign economy. The government determines the path of public debt, while the central bank sets the stance of monetary policy.

In the first part of this section, we model the international financial shock as a wedge in the Backus–Smith condition of international risk sharing. This modeling specification is particularly convenient because it yields closed-form solutions for nominal and real variables, allowing for a transparent characterization of the underlying mechanisms. In the second part of the section, the international financial shock takes the well-known form of a currency demand shock in a model with incomplete financial markets à la Gabaix and Maggiori (2015). In this case, a closed-form solution is not available. However, we compare the numerical simulations of the two specifications to show that they are quantitatively equivalent in Table C.1 of the Appendix.

## 2.1 Households

### 2.1.1 Intratemporal problem

The consumption bundle $C_t$ is a constant elasticity of substitution (CES) aggregator of home and foreign goods:

$$C_t = \left[(1-\gamma)^{\frac{1}{\theta}} C_{Ht}^{\frac{\theta-1}{\theta}} + \gamma^{\frac{1}{\theta}} C_{Ft}^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \tag{1}$$

where $\theta > 0$ is the elasticity of intratemporal substitution, and $C_{Ht}$ and $C_{Ft}$ are the consumption of domestic and foreign goods, respectively:

$$C_{Ht} = (1-\gamma)\left(\frac{P_{Ht}^N}{P_t}\right)^{-\theta} C_t \tag{2}$$

$$C_{Ft} = \gamma\left(\frac{E_t P_t^*}{P_t}\right)^{-\theta} C_t. \tag{3}$$

Here, $P_{Ht}^N$ is the home-currency price of the domestic good; $P_t$ and $P_t^*$ are the price indexes of home and foreign consumption, respectively; $C_t^*$ denotes exogenous foreign consumption; and $E_t$ is the nominal exchange rate, defined as the price of one unit of foreign currency in terms of the home currency.[1]

[1]Since the home country is modeled as a small open economy, the foreign price index coincides with the price of the foreign good.

The CES aggregator implies the following price index of home consumption:

$$P_t = \left[ (1-\gamma) P_{Ht}^{N1-\theta} + \gamma \left( E_t P_t^* \right)^{1-\theta} \right]^{\frac{1}{1-\theta}}. \tag{4}$$

Dividing both sides by $P_t$, we get:

$$1 = (1-\gamma) P_{Ht}^{1-\theta} + \gamma Q_t^{1-\theta}, \tag{5}$$

where $P_{Ht} \equiv \frac{P_{Ht}^N}{P_t}$ and $Q_t \equiv \frac{E_t P_t^*}{P_t}$ is the real exchange rate, which can be expressed solely in terms of stationary variables:

$$\Delta Q_t = \Delta E_t \frac{\Pi_t^*}{\Pi_t}. \tag{6}$$

Here, $\Pi_t$ and $\Pi_t^*$ are consumer-price-index gross inflation, $\Delta Q_t \equiv \frac{Q_t}{Q_{t-1}}$, and $\Delta E_t \equiv \frac{E_t}{E_{t-1}}$.

### 2.1.2 Intertemporal problem

The representative household maximizes a discounted expected utility over final consumption $C_t$ and labor $L_t$:

$$\max_{\{C_t, L_t, B_t, B_{Gt}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left( \frac{1}{1-\sigma} C_t^{1-\sigma} - \frac{1}{1+\frac{1}{\nu}} L_t^{1+\frac{1}{\nu}} \right),$$

where $0 < \beta < 1$ is the household discount factor; $\sigma > 0$ is the relative-risk-aversion parameter; and $\nu$ is the Frisch elasticity of labor supply. The maximization problem is subject to the following budget constraint:

$$C_t + \sum_{s_{t+1} \in S} \mathcal{S}_t \left( s_{t+1} \right) B_{t+1} \left( s_{t+1} \right) + B_{Gt} \leq W_t L_t + \frac{B_t}{\Pi_t} + \frac{R_{t-1}}{\Pi_t} B_{Gt-1} + Prof_t - T_t. \tag{7}$$

Here, $B_{t+1}\left(s_{t+1}\right)$ denotes the amount of state-contingent bonds purchased at time $t$ and delivering one unit of home currency in period $t+1$ under state $s_{t+1}$, with $S$ the set of possible states; $\mathcal{S}_t\left(s_{t+1}\right)$ is the price of the state-contingent bond; $B_{Gt}$ denotes the real holdings of government bonds, which yield the nominal rate $R_t$; $W_t$ is the real wage; $Prof_t$ denotes the profits of the firms; and $T_t$ denotes lump-sum taxes.

The first-order conditions of the household problem yield a labor supply expression:

$$L_t^{\frac{1}{\nu}} = W_t C_t^{-\sigma}; \tag{8}$$

and a Euler equation for bonds:

$$1 = \beta \mathbb{E}_t \left[ \left( \frac{C_t}{C_{t+1}} \right)^{\sigma} \frac{R_t}{\Pi_{t+1}} \right]. \tag{9}$$

We define the real interest rate with $RR_t$:

$$RR_t = \frac{R_t}{\mathbb{E}_t\left(\Pi_{t+1}\right)}. \tag{10}$$

We then impose a modified Backus-Smith condition:

$$\left(\frac{C_t e^{\psi_t}}{C_t^*}\right)^{\sigma} = Q_t, \tag{11}$$

where $\psi_t$ is a reduced-form wedge that breaks international risk sharing and follows an AR(1) process:

$$\psi_t = \rho_\psi \psi_{t-1} + \varepsilon_t^\psi, \tag{12}$$

with $\rho_\psi \in [0, 1)$ and $\varepsilon_t^\psi$ an exogenous shock. This shock plays a role analogous to that of international financial flows in Gabaix and Maggiori (2015) and to the shocks generated by noisy traders in Itskhoki and Mukhin (2021, 2025). In these frameworks, the shock takes the form of a currency demand shock and operates only when the nominal exchange rate is not perfectly pegged; otherwise, the carry trade return is riskless and the UIP condition holds exactly. Note that, all else equal, a positive financial shock reduces consumption and depreciates the nominal and real exchange rates, exactly as in the papers cited above. Here, we simply append the shock to the Backus-Smith condition in order to derive a closed-form solution. We provide a microfoundation for the wedge in Subsection 2.7 and show, by simulating the model, that the results are qualitatively and quantitatively similar to those obtained with the closed-form solution.

## 2.2 Firms

The representative firm produces the domestic good $Y_{Ht}$ using the following production function, under perfect competition and producer currency pricing:

$$Y_{Ht} = e^{a_t} L_t, \tag{13}$$

where $a_t$ denotes the total factor productivity (TFP), which follows an AR(1) process:

$$a_t = \rho_a a_{t-1} + \epsilon_t^a. \tag{14}$$

Here, $\epsilon_t^a$ is a TFP shock and $\rho_a \in [0, 1)$. The profit maximization problem yields the following condition, equating the price to the real marginal cost of production:

$$P_{Ht} = \frac{W_t}{e^{a_t}}. \tag{15}$$

## 2.3 Policy

The government budget constraint reads:

$$D_t + \frac{R_{t-1}}{\Pi_t} B_{Gt-1} = B_{Gt}. \tag{16}$$

Here, the left-hand side includes public expenditures, comprising the primary deficit $D_t$ and the interest on the public debt; the right-hand side represents revenues from new public debt issuance. We assume that public spending on consumption is zero, so $D_t = -T_t$.

The public sector, which comprises the government and the central bank, controls two policy instruments: the deficit and the nominal interest rate. We consider three dummy variables that capture the three policy regimes—*floating*, *pegged*, and *fiscal*—which are summarized by the following policy rules:

$$R_t = R \left( \frac{\Pi_t}{\bar{\Pi}} \right)^{floating \cdot \phi_\pi} \left( \frac{\Delta E_t}{\Delta \bar{E}} \right)^{pegged \cdot \phi_e} \tag{17}$$

$$D_t = \bar{D} + \left[ - \left( floating + pegged \right) \phi_b \left( B_{Gt-1} - \bar{B_G} \right) + fiscal \cdot \rho_d \left( D_{t-1} - \bar{D} \right) \right] + v_t^d, \tag{18}$$

where $\bar{\Pi}$ is the target for gross inflation; $\Delta \bar{E}$ is the target for the change in the nominal exchange rate; $\bar{D}$ is the target for primary deficit; $\bar{B_G}$ is the target for public debt; and $v_t^d$ is an exogenous shock, which we label fiscal shock.[2]

Under the *floating* regime, the central bank uses a Taylor rule with $\phi_\pi > 1$ to smooth inflation fluctuations around the target. Assuming $-1 + \beta^{-1} < \phi_b < 1 + \beta^{-1}$, fiscal policy is passive: public debt converges back to its steady-state level, ensuring that fiscal shocks are funded by future primary surpluses.[3] Under the *pegged* regime, the government uses the same deficit rule as in the *floating* regime. Assuming $\phi_e \to \infty$, the exchange-rate depreciation is constant.[4] In these two regimes, monetary policy is active and fiscal policy is passive, in the sense of Leeper (1991).

In the *fiscal* regime, fiscal dominance prevails: monetary policy is passive, since the central bank sets the nominal interest rate at a constant level, while fiscal policy is active, as equation (18) becomes an AR(1) process for the public deficit, implying that fiscal shocks are unfunded. Remarkably, in Subsection 3.3, we show that the same results are obtained when the central bank expands the money supply to finance the unfunded fiscal shock.

[2] Variables without a time index are understood to be in steady state. Since, in steady state, gross domestic product (GDP) is normalized to one, $D_t$ can be interpreted as the primary deficit as a share of steady-state GDP.

[3] The lower bound requires the government to reduce the deficit by more than the additional interest payments generated by higher debt, since $-1 + \beta^{-1}$ is the steady-state real interest rate. The upper bound rules out an excessively strong fiscal correction, which would push debt beyond its steady state in the opposite direction every period, generating oscillations of increasing amplitude.

[4] If the home inflation target differs from the foreign inflation target, the nominal exchange rate changes at a constant rate $\bar{\Delta E} \neq 1$ in steady state, as implied by equation (6).

## 2.4 Market clearing

The market clearing condition for domestic goods is as follows:

$$Y_{Ht} = (1-\gamma)\left(P_{Ht}\right)^{-\theta} C_t + \gamma \left(\frac{P_{Ht}}{Q_t}\right)^{-\theta} C_t^*. \tag{19}$$

The domestic good is consumed by home households—the first term on the right-hand side—and by foreign households—the second term on the right-hand side, which is the foreign counterpart of equation (3).

Combining equations (8), (13), and (15), we obtain the equilibrium in the labor market:

$$Y_{Ht}^{\frac{1}{\nu}} C_t^{\sigma} = P_{Ht} e^{a_t\left(1+\frac{1}{\nu}\right)}. \tag{20}$$

Finally, GDP is defined as:

$$GDP_t = P_{Ht} Y_{Ht}. \tag{21}$$

## 2.5 Equilibrium

Our economy can be divided into two blocks. The first block (henceforth, the real block) includes five endogenous variables $\{C_t, Y_{Ht}, P_{Ht}, Q_t, RR_t\}$, whose equilibrium is described by equations (5), (9), (11), (19), and (20), and by the exogenous processes for $a_t$ and $\psi_t$, assuming constant foreign variables ($C_t^* = \Pi_t^* = 1$). This block is independent of the policy regime, with one important exception: under the *pegged* regime, international financial shocks are absent.

The second block (henceforth, the nominal block) includes five endogenous variables $\{B_{Gt}, \Delta E_t, R_t, \Pi_t, D_t\}$, whose equilibrium is described by equations (6), (10), (16), (17), and (18), given the evolution of the variables in the real block. The separation into two blocks reflects the monetary dichotomy: under flexible prices and in the absence of monetary frictions, real variables are independent of nominal variables.

## 2.6 Analytical results

We linearize the model around the non-stochastic steady state. Lowercase variables denote percentage deviations from their steady-state values, except for $b_{Gt}$ and $d_t$, which denote deviations from steady state expressed as a share of steady-state GDP (normalized to one).

After eliminating $p_{Ht}$ using (5) and linearizing the equations (9), (11), (19), and (20), the real block becomes:

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma} rr_t \tag{22}$$

$$y_{Ht} = (1-\gamma)\, c_t + \theta\gamma \frac{2-\gamma}{1-\gamma} q_t \tag{23}$$

$$\frac{1}{\nu} y_{Ht} + \sigma c_t = -\frac{\gamma}{1-\gamma} q_t + \left(1 + \frac{1}{\nu}\right) a_t \tag{24}$$

$$c_t = \frac{1}{\sigma} q_t - \psi_t. \tag{25}$$

Equation (22) is the linearized Euler equation that links consumption growth with the real interest rate. Equation (23) is the market-clearing condition, which equates domestic production with home and foreign demand, the latter depending on the real exchange rate. Equation (24) represents the equilibrium in the labor market. Equation (25) represents the Backus–Smith condition augmented by international financial shocks.

Note that, in all regimes, the real block is unaffected by fiscal shocks and responds only to TFP shocks and international financial shocks. The solution to the model is given by (see Appendix A):

$$c_t = \frac{(1+\nu)(1-\gamma)}{\omega + \sigma\nu} a_t - \gamma\sigma \frac{\nu + \theta(2-\gamma)}{\omega + \sigma\nu} \psi_t \tag{26}$$

$$y_{Ht} = \omega \frac{1+\nu}{\omega + \sigma\nu} a_t + \gamma\sigma\nu \frac{\theta\sigma(2-\gamma) - (1-\gamma)}{\omega + \sigma\nu} \psi_t \tag{27}$$

$$rr_t = -\Psi_a a_t + \Psi_\psi \psi_t \tag{28}$$

$$q_t = \Xi_a a_t + \Xi_\psi \psi_t, \tag{29}$$

Here, $\omega \equiv (1-\gamma)^2 + \theta\gamma\sigma(2-\gamma)$, $\Psi_a \equiv \frac{\sigma(1-\rho_a)(1+\nu)(1-\gamma)}{\omega+\sigma\nu}$, $\Psi_\psi \equiv \frac{\gamma(1-\rho_\psi)\sigma^2[\nu+\theta(2-\gamma)]}{\omega+\sigma\nu}$, $\Xi_\psi \equiv \sigma(1-\gamma)\frac{1-\gamma+\sigma\nu}{\omega+\sigma\nu}$, and $\Xi_a \equiv \sigma\frac{(1+\nu)(1-\gamma)}{\omega+\sigma\nu}$ are all positive combinations of parameters. This solution applies to all three regimes, except that international financial shocks are absent under the *pegged* regime.

The dynamics following a positive TFP shock are as expected: output increases and, as the supply of the domestic good expands, the real exchange rate depreciates. Consumption also increases, while households save in order to smooth the shock, thereby putting downward pressure on the real interest rate.

A positive international financial shock depreciates the real exchange rate and reduces consumption through the Backus–Smith condition. Output is affected by two opposing forces. On the one hand, the depreciation of the real exchange rate increases foreign demand. On the other hand, the decline in consumption reduces domestic demand. The former effect dominates when $\theta$ is sufficiently high, i.e, when foreign demand is sufficiently elastic with respect to the real exchange rate. The real interest rate increases because, given the transitory nature of the shock, households anticipate higher consumption in the future relative to today.

### 2.6.1 *Floating* regime

Inflation is jointly determined by combining the monetary rule with equation (10):

$$\pi_t = -\frac{\Psi_a}{\phi_\pi - \rho_a} a_t + \frac{\Psi_\psi}{\phi_\pi - \rho_\psi} \psi_t, \tag{30}$$

under the assumption that $\phi_\pi > 1$ (see Galí, 2015). This result also yields the solution for the nominal interest rate $r_t = \phi_\pi \pi_t$.

Using equation (6), we find the solution for the nominal exchange rate:

$$\Delta e_t = -\frac{\Psi_a}{\phi_\pi - \rho_a} a_t + \Xi_a \left(a_t - a_{t-1}\right) + \frac{\Psi_\psi}{\phi_\pi - \rho_\psi} \psi_t + \Xi_\psi \left(\psi_t - \psi_{t-1}\right). \tag{31}$$

The first term captures the decline in inflation following a positive TFP shock, which puts appreciation pressure on the nominal exchange rate. The second term captures the real depreciation induced by the shock, which puts depreciation pressure on the nominal exchange rate. Using $\Psi_a = (1 - \rho_a)\, \Xi_a$, we can rewrite the nominal depreciation equation as follows:

$$\Delta e_t = \frac{\phi_\pi - 1}{\phi_\pi - \rho_a} \Xi_a a_t - \Xi_a a_{t-1} + \frac{\Psi_\psi}{\phi_\pi - \rho_\psi} \psi_t + \Xi_\psi \left(\psi_t - \psi_{t-1}\right) \tag{32}$$

Under the maintained assumptions $\phi_\pi > 1$ and $\rho_a < 1$, the impact coefficient on a positive TFP innovation is strictly positive. The real-depreciation channel therefore dominates on impact, and the nominal exchange rate depreciates.

A positive international financial shock raises inflation and, together with the resulting real depreciation, leads to a depreciation of the nominal exchange rate. For a sufficiently high Taylor parameter $\phi_\pi$, the nominal and real exchange rates move in tandem in response to international financial shocks.

By contrast, fiscal shocks leave inflation and the nominal exchange rate unaffected in the *floating* regime. Their only effects are on public debt and the deficit, whose dynamics are described by the following two equations:

$$b_{Gt} = d_t + \frac{1}{\beta} b_{Gt-1} + \frac{B_G}{\beta} \left(r_{t-1} - \pi_t\right) \tag{33}$$

$$d_t = -\phi_b b_{Gt-1} + v_t^d. \tag{34}$$

The fiscal shock increases the current deficit, which in turn raises the stock of public debt. If $\phi_b > 0$, fiscal policy is passive, as the government reduces future deficits to repay the debt generated by the fiscal shock, thereby ensuring that the shock is funded.

### 2.6.2 *Pegged* regime

In the *pegged* regime, the nominal exchange rate is, by assumption, unaffected by any shock:

$$\Delta e_t = 0. \tag{35}$$

By equation (6), inflation is given by:

$$\pi_t = -\Xi_a \left(a_t - a_{t-1}\right), \tag{36}$$

which decreases on impact following the real depreciation. The nominal interest rate is given by equation (10):

$$r_t = \left[-\Psi_a + (1-\rho_a)\,\Xi_a\right] a_t = 0, \tag{37}$$

and therefore remains constant. As in the *floating* regime, deficit shocks do not affect nominal variables, and equations (33)-(34) fully characterize the dynamics of $b_{Gt}$ and $d_t$.

### 2.6.3 *Fiscal* regime

The deficit follows a purely exogenous process driven by unfunded fiscal shocks:

$$d_t = \rho_d d_{t-1} + v_t^d. \tag{38}$$

Using $r_t = 0$, inflation and public debt are jointly determined by equations (10) and (16), whose log-linearized versions are the following:

$$\mathbb{E}_t \pi_{t+1} = -r r_t \tag{39}$$

$$\pi_t = \frac{1}{B_G}\left[b_{Gt-1} + \beta\left(d_t - b_{Gt}\right)\right]. \tag{40}$$

Solving the system, we get:

$$b_{Gt} = \frac{B_G \Psi_a}{1-\beta\rho_a} a_t - \frac{B_G \Psi_\psi}{1-\beta\rho_\psi}\psi_t - \frac{\beta\rho_d}{1-\beta\rho_d} d_t, \tag{41}$$

$$\pi_t = -\frac{\Psi_a}{1-\beta\rho_a}\left(\beta a_t - a_{t-1}\right) + \frac{\Psi_\psi}{1-\beta\rho_\psi}\left(\beta\psi_t - \psi_{t-1}\right) + \frac{\beta}{B_G\left(1-\beta\rho_d\right)}\left(d_t - \rho_d d_{t-1}\right). \tag{42}$$

Inflation still responds to TFP shocks and international financial shocks. On impact, a positive TFP shock reduces inflation, whereas a positive financial shock raises it, through a different mechanism than in the other two regimes.

A positive TFP shock reduces the real interest rate, requiring lower current inflation to compensate for the resulting reduction in debt service. The opposite occurs following a positive international financial shock.

Unfunded fiscal shocks affect inflation through the mechanism at the heart of the fiscal theory of the price level. A higher deficit is not financed by future surpluses, but instead by reducing the real value of public debt through inflation. Instead, expected inflation remains pinned down by the current real interest rate, and only inflation is affected by an unexpected unfunded fiscal shock $v_t^d = d_t - \rho_d d_{t-1}$.

Finally, we are ready to derive a closed-form solution for the nominal exchange rate under fiscal dominance, which is the main contribution of our analysis within the simple model. Using equation (6), we find that:

$$\begin{aligned}\Delta e_t =& \Xi_a \left(a_t - a_{t-1}\right) - \frac{\Psi_a}{1-\beta\rho_a}\left(\beta a_t - a_{t-1}\right) + \\ &+ \frac{\Psi_\psi}{1-\beta\rho_\psi}\left(\beta\psi_t - \psi_{t-1}\right) + \Xi_\psi\left(\psi_t - \psi_{t-1}\right) + \\ &+ \frac{\beta}{B_G\left(1-\beta\rho_d\right)}\left(d_t - \rho_d d_{t-1}\right), \end{aligned} \tag{43}$$

implying that the nominal exchange rate also depends on the unfunded fiscal shock.

A positive unfunded fiscal shock raises inflation and transmits one-for-one to the nominal exchange rate. Two features are worth noting. First, as in the case of inflation, only unexpected fiscal shocks affect the nominal exchange rate: expected fiscal shocks do not affect inflation and therefore do not transmit to the nominal exchange rate. Second, the lower the level of public debt, the stronger the response of the nominal exchange rate, as a smaller debt base requires a larger increase in inflation to absorb a given unfunded fiscal shock.

The introduction of unfunded fiscal shocks represents the key innovation of our framework. These shocks weaken the comovement between nominal and real exchange rates, generating substantial volatility in inflation and the nominal exchange rate without affecting the real exchange rate. In contrast, TFP shocks and international financial shocks cannot weaken this comovement, as they move the nominal and real exchange rates in the same direction when monetary policy stabilizes inflation.[5]

We also consider a version of our quantitative model with a binding constraint on public debt. In this specification, fiscal shocks induce endogenous adjustments in money creation and inflation consistent with the government budget constraint and the public debt limit. The main results remain qualitatively unchanged: inflation is volatile, and the comovement between nominal and real exchange rates weakens. This exercise shows that the mechanism also operates under monetary accommodation of fiscal financing needs (Sargent 2013).

[5] As shown by Itskhoki and Mukhin (2025), if international financial shocks are sufficiently volatile in a floating exchange-rate regime and are therefore the primary drivers of nominal and real exchange rates, switching from a floating to a pegged exchange-rate regime generates a sharp reduction in the volatility of both nominal and real exchange rates. In a pegged exchange-rate regime, the volatility of $\Delta e_t$ is zero (or low) by assumption, while the volatility of $\Delta q_t$ is substantially lower than in the floating regime, as international financial shocks are absent. This mechanism replicates the classic Mussa puzzle but cannot account for the weaker comovement between nominal and real exchange rates observed in Brazil from the 1980s through mid-1994.

## 2.7 Incomplete financial markets

We now augment the simple model with incomplete financial markets, rather than modeling the international financial shock as a wedge in the Backus–Smith condition for international risk sharing, while leaving the remainder of the model unchanged. We follow Gabaix and Maggiori (2015) and assume that households can only trade home bonds $B_t$, intermediated by a representative financier owned by foreign households.

In period $t$ the financier buys $F_t^*$ foreign bonds and sells $F_t$ home bonds, operating without capital:

$$\frac{F_t}{Q_t} + F_t^* = 0. \tag{44}$$

In period $t+1$, the financier's profits, $\Omega_{t+1}$, denominated in foreign currency, are given by:

$$\Omega_{t+1} = \left( \frac{R_t^*}{\Pi_{t+1}^*} - \frac{R_t}{\Pi_{t+1}} \frac{Q_t}{Q_{t+1}} \right) F_t^*.$$

Here, $RR_t^*$ is the real foreign interest rate. Following Gabaix and Maggiori (2015), the intermediary can divert a portion $\Gamma\left[var(\Delta e)\right]|F_t^*|$ of the position $F_t^*$, where the function $\Gamma\left[var(\Delta e)\right]$ measures the shallowness of international financial markets, with $\Gamma(0) = 0$. Its argument is the unconditional variance of the nominal exchange rate and, following Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2025), differs between exchange-rate regimes.

The bank is subject to the following constraint:

$$V_t \geq \Gamma |F_t^*| |F_t^*| = \Gamma F_t^{*2}, \tag{45}$$

where $V_t = \max_{F_t^*} \beta \mathbb{E}_t \left[ \left( \frac{C_t^*}{C_{t+1}^*} \right)^\sigma \Omega_{t+1} \right]$ is the value of the intermediary. As shown by Gabaix and Maggiori (2015), the constraint is always binding, and we obtain:

$$\Gamma F_t^* = \beta \mathbb{E}_t \left[ \left( \frac{C_t^*}{C_{t+1}^*} \right)^\sigma \left( RR_t^* - RR_t \frac{Q_t}{Q_{t+1}} \right) \right], \tag{46}$$

which is a modified UIP condition. Remarkably, the UIP condition holds exactly when $\Gamma = 0$.

Finally, we assume that noisy traders in the foreign country trade bonds without capital to introduce a shock into the UIP condition:

$$N_t^* + \frac{N_t}{Q_t} = 0, \tag{47}$$

where $N_t$ and $N_t^*$ denote the positions taken by noisy traders in home and foreign bonds, respectively, and are modeled as exogenous processes:

$$N_t^* = e^{\psi_t} - 1. \tag{48}$$

Here,

$$\psi_t = \rho_\psi \psi_{t-1} + \varepsilon_t^\psi, \tag{49}$$

where $\varepsilon_t^\psi$ is an international financial shock, and $\rho_\psi \in [0, 1)$. In equilibrium, the following must hold:

$$nB_t + (1-n)(N_t + F_t) = 0 \tag{50}$$

$$(1-n)(B_t^* + N_t^* + F_t^*) = 0, \tag{51}$$

where $B_t^*$ denotes the position taken by foreign households in foreign bonds and $n$ is the home's share of the world population (assumed to be close to zero). Using equations (44)-(47), combined with the previous market clearing conditions, we get:

$$B_t = -\frac{1-n}{n} Q_t B_t^*. \tag{52}$$

We can then rewrite the UIP condition as follows:

$$\frac{B_t}{Q_t} = \frac{1-n}{\Gamma n}\beta \left\{ \mathbb{E}_t \left[ \left( \frac{C_t^*}{C_{t+1}^*} \right)^\sigma \left( RR_t^* - RR_t \frac{Q_t}{Q_{t+1}} \right) \right] \right\} + \frac{1-n}{n} N_t^*. \tag{53}$$

Taking a first-order Taylor approximation of the previous equation around a steady state with $B = 0$ and assuming a constant foreign rate, we get:

$$rr_t = \mathbb{E}_t q_{t+1} - q_t + \Gamma \psi_t - \iota b_t. \tag{54}$$

Here, $\iota \equiv \frac{\Gamma n}{1-n}$ and $b_t = \frac{B_t - B}{Y}$.[6] This equation is equivalent to the log-linearized UIP condition in Itskhoki and Mukhin (2025) and to similar expressions obtained assuming bond adjustment costs rather than financial market segmentation (see Yakhin, 2022). Equation (54) replaces equation (25) in the analytical model.

We need to specify $\Gamma$ as a function of nominal exchange-rate volatility. For simplicity, we assume that $\Gamma$ takes one of two values, depending on the exchange-rate policy: a lower value when the nominal exchange rate is pegged (equal to zero if the peg is perfect) and a higher value when it is not. Note that there is an additional variable, $b_t$, representing the bond holdings of home households, which coincide in equilibrium with the home net foreign asset position.

[6] In Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2025), the UIP condition holds exactly, and the foreign bond position $B_t$ drops out when the exchange rate is perfectly pegged. Here, we assume that $B_t$ continues to enter the UIP condition under the *pegged* regime in order to preserve stationarity (see Schmitt-Grohé and Uribe, 2003).

The additional equilibrium condition is the resource constraint of the home economy:

$$C_t + B_t - \frac{B_{t-1}R_{t-1}}{\Pi_t} = P_{Ht}Y_{Ht}, \tag{55}$$

which, up to a first-order approximation, becomes:

$$c_t + b_t = \frac{1}{\beta}b_{t-1} - \frac{\gamma}{1-\gamma}q_t + y_{Ht}. \tag{56}$$

The presence of the state variable $b_t$ precludes a closed-form solution. We therefore resort to numerical simulations of the two specifications of the simple model to show that they are quantitatively equivalent (Table C.1 of the Appendix).

## 2.8 Monetary–Fiscal policy interactions: evidence from Brazil (1980-2019)

The main purpose of this subsection is to present a set of stylized facts—some classic and others new—that emerge from an empirical analysis of Brazilian exchange rates over the period 1980–2019. These include (i) the classic Mussa puzzle: a change in the exchange-rate regime that increases (decreases) the volatility of the nominal exchange rate also increases (decreases) the volatility of the real exchange rate; (ii) the well-known disconnect between exchange rates, both nominal and real, when the fiscal authority stabilizes debt; the reconnection between the nominal exchange rate and inflation—accompanied by a weakening of the comovement between nominal and real exchange rates, when the fiscal authority does not stabilize debt. This last stylized fact has been less explored by the literature.

Figure 1 displays the first difference of the logarithm of the bilateral nominal exchange rate (in blue) and the first difference of the logarithm of the bilateral real exchange rate (in light red) between Brazil and the United States during the period 1980–2019. Both series exhibit two structural breaks in volatility: the introduction of the peg to the US dollar in July 1994, associated with the implementation of the *Plano Real*, and the collapse of the peg in January 1999. These breaks are consistent with the classic Mussa puzzle. However, this is not the full story.

Although the periods from January 1980 to July 1994 and from January 1999 onward both fall under floating exchange-rate regimes, they differ markedly. In particular, the annual inflation rate averaged 353.25% from January 1980 to July 1994 and 6.33% from January 1999 to December 2019, and the comovement between nominal and real exchange rates is substantially weaker in the earlier period than in the later period. This point is illustrated in Figure 2, which consists of two panels.

The top panel of Figure 2 displays the conditional variances of the nominal and real exchange rates (shown by the blue and red lines, respectively), estimated using a simple GARCH model of order (1,1). From July 1994 onward, these variances comove closely, whereas during the period from January 1980 to July 1994 the conditional variance of the nominal exchange rate is significantly higher than that of the real exchange rate.

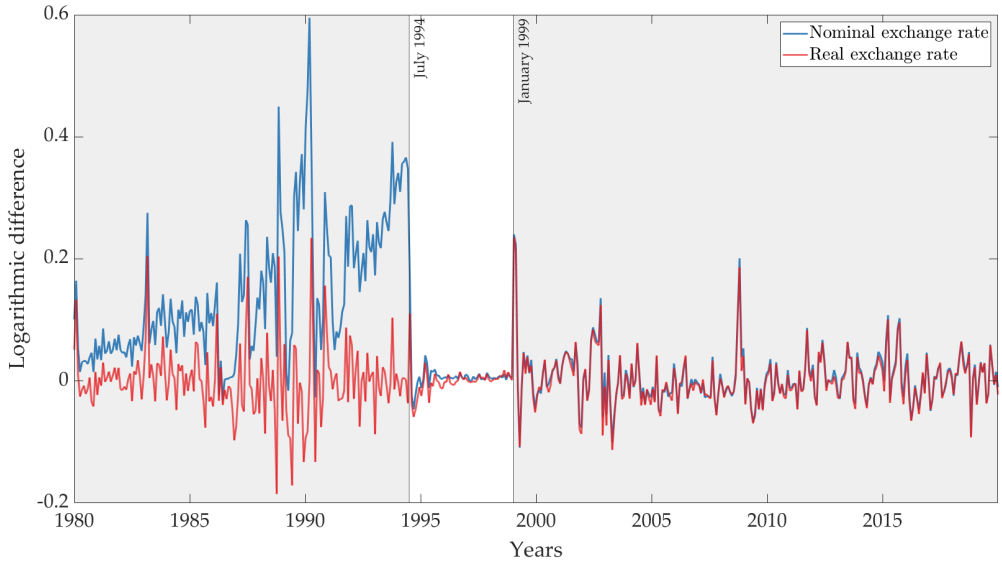



Figure 1: Exchange rates for Brazil (home country) and United States (foreign country)

*Notes:* The exchange-rate series are expressed in logarithmic differences. The vertical lines indicate shifts in the exchange-rate regime. Periods under pegged exchange-rate regimes are unshaded, whereas periods under floating exchange-rate regimes are shaded.
*Sources:* The Exchange Rates Portal (Bank of Italy) and the International Financial Statistics (International Monetary Fund).

The bottom panel of Figure 2 instead compares the conditional variance of the inflation difference between Brazil and the United States (green line), also estimated using a GARCH(1,1) model, with that of the nominal exchange rate. It shows that (i) the conditional variance of the inflation difference is much higher prior to July 1994, and (ii) its comovement with the nominal exchange rate is strong before July 1994 but considerably weaker thereafter.

This evidence points to two novel insights that provide a natural setting for testing our theoretical framework. First, factors driving the Brazilian inflation before July 1994 also appear to drive the nominal exchange rate. Second, the *Plano Real*, which introduced a pegged exchange-rate regime, reduced not only the high conditional variance of the nominal exchange rate but also that of the inflation difference. Indeed, an examination of the monetary and fiscal aggregates before and after July 1994 in Kehoe and Nicolini (2022) further indicates that the implementation of the *Plano Real* represented not only a change in Brazilian exchange-rate policy but also a shift in fiscal policy, from a regime of fiscal dominance to one of monetary dominance. In particular, the average annual total deficit (as a percentage of GDP) decreased from 5.8% between 1990 and 1993 to 1.2% during the period 1994–1998. We explicitly incorporate these two insights into our model through the role of unfunded fiscal shocks in the *fiscal* regime and the explicit interaction between monetary and fiscal policies, which allows us to replicate the three periods of the Brazilian economy: January 1980–June 1994, July 1994–December 1998, and January 1999–December 2019.[7]

[7]In our analysis, we characterize the post-1999 period as one of monetary dominance and a floating exchange-rate regime in Brazil. However, some authors (e.g., Alberola et al. 2022) correctly argue that not all fiscal shocks are funded after 1999, in the sense of Bianchi, Faccini, and Melosi (2023). Nevertheless, we retain the assumption of monetary dominance both to make the distinction between the pre-1994 and post-1999 periods more transparent and because there have been no further changes in Brazil's exchange-rate regime since 1999.

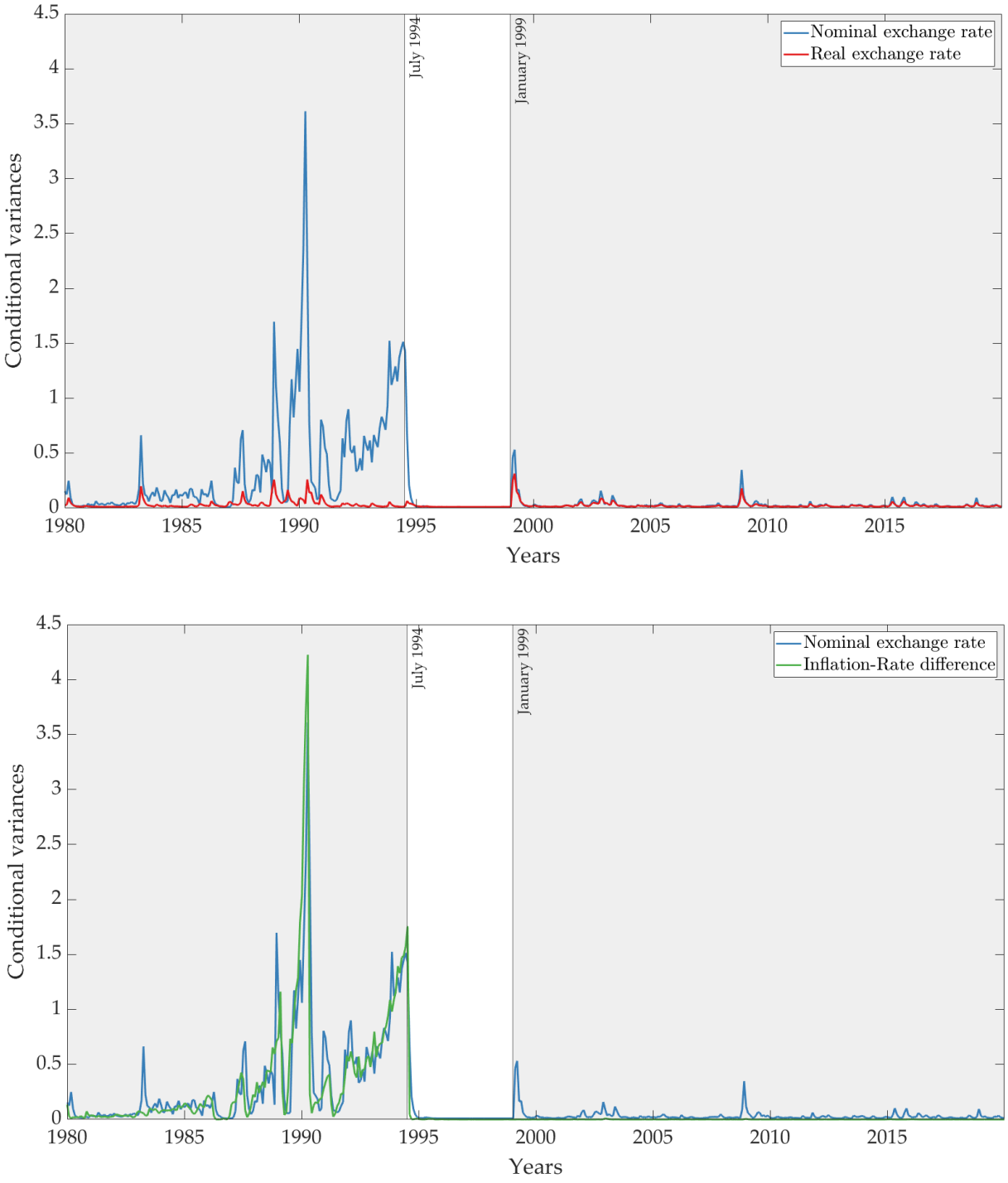



Figure 2: Conditional variances for exchange rates and inflation-rate difference

*Notes for top panel:* The series of the conditional variances are annualized. The vertical lines indicate shifts in the exchange-rate regime. Periods under pegged exchange-rate regimes are unshaded, whereas periods under floating exchange-rate regimes are shaded.
*Notes for bottom panel:* The series of the conditional variances are annualized. The vertical lines indicate shifts in the exchange-rate regime. Periods under pegged exchange-rate regimes are unshaded, whereas periods under floating exchange-rate regimes are shaded.
*Sources:* The Exchange Rates Portal (Bank of Italy) and the International Financial Statistics (International Monetary Fund).

## 2.9 Simulation

We simulate the simple model to show that, despite its simplicity, it can account for several empirical facts. We report three sets of results. First, we compare selected moments from the model with their empirical counterparts. Second, we report the variance decomposition of selected variables in each policy regime. Third, we plot the corresponding impulse response functions.

### 2.9.1 Calibration

We use a quarterly calibration, summarized in Table 1. The sample periods for the three regimes are 1980Q2—1994Q2 for the *fiscal* regime, 1994Q3—1998Q4 for the *pegged* regime, and 1999Q1—2019Q4 for the *floating* regime.

Following Itskhoki and Mukhin (2025), we set $\beta = 0.99$, $\sigma = 2$, $\nu = 1$, $\theta = 1.5$, $\iota = 0.001$, and $\phi_\pi = 2.15$, set all autoregressive coefficients to $0.97$, normalize the standard deviation of TFP shocks to one, and calibrate the standard deviation of financial shocks to $0.14$ to match a correlation of $-0.25$ between Brazilian consumption growth and the growth rate of the Brazilian real exchange rate vis-à-vis the G7 economies over the period 1999Q1–2019Q4.[8].

We set the trade openness $\gamma$ to $0.122$ to match the Brazilian import-to-GDP ratio of $12.2\%$. Steady-state inflation is set to $\Pi = 1 + \frac{0.03}{4}$, consistent with Brazil's $3\%$ inflation target.

Finally, some parameters are relevant only under the *fiscal* regime. We set the standard deviation of deficit shocks to $0.204$ to match the standard deviation of inflation relative to the output growth of $4.78$ observed in Brazil during 1980-1994. We set the steady-state public debt-to-GDP ratio to $40.25\%$, corresponding to the Brazilian average over the same period, and set $\phi_b = 0.09$ following the Bayesian estimates of Bianchi, Faccini, and Melosi (2023).

### 2.9.2 Moments and variance decompositions

Table 2 shows that our model replicates the three regimes of the Brazilian economy shown in Figure 1. The first column refers to the simple model, presented in this section; the second column refers to the quantitative model, presented in Section 3; the third column reports the empirical moments.

In the *floating* regime, the correlation between the nominal and the real exchange rate is virtually 1, consistent with the data for the period since 1999Q1 (Figure 1). The model captures the weak correlation between the nominal exchange rate and inflation, as well as the fact that the nominal exchange rate is an order of magnitude more volatile than inflation. It also closely matches the relative volatility of inflation to output and captures the higher volatility of the real exchange rate relative to output, although its volatility is only about half that observed in the data.

[8]Itskhoki and Mukhin (2025) target the Backus-Smith correlation, i.e., the correlation between the real exchange rate and the home-foreign consumption growth differential, which is $-0.22$ in our sample. Since foreign consumption is constant in our model, we instead target the correlation between the real exchange rate and home consumption growth, which is $-0.25$.

Table 1: Calibrated parameters

| Parameter | Description | Value | Parameter | Description | Value |
|---|---|---|---|---|---|
| $\beta$ | Discount factor | 0.99 | $\phi_\pi$ | Taylor coefficient | 2.15 |
| $\sigma$ | Relative risk aversion | 2 | $\phi_b$ | Fiscal rule coefficient | 0.09 |
| $\nu$ | Frisch elasticity | 1 | $\Pi$ | Inflation target | 1.0075 |
| $\gamma$ | Trade openness | 0.122 | $B_G$ | SS public debt | 1.61 |
| $\theta$ | Elasticity of substitution | 1.5 | $\sigma_a$ | SD of TFP shock | 1 |
| $\iota$ | Shallowness of int. fin. markets | 0.001 | $\Gamma \cdot \sigma_\psi$ | SD of financial shock | 0.14 |
| $\rho_a, \rho_\psi, \rho_d$ | AR(1) parameter | 0.97 | $\sigma_d$ | SD of fiscal shock | 0.2043 |

In the *pegged* regime, the volatility of the nominal exchange rate is zero by construction, while the absence of international financial shocks reduces the volatility of the real exchange rate relative to the *floating* regime, consistent with the classic Mussa puzzle.[9]

The *fiscal* regime is the main contribution of our work: unfunded fiscal shocks move inflation and nominal exchange-rate depreciation in the same direction without affecting the real exchange rate. These shocks therefore explain why the nominal and real exchange rates are much less correlated in Brazil before 1994Q2. The *fiscal* regime also captures other moments well, including the volatility of the nominal and real exchange rates.

Table 3 reports the variance decomposition. The fiscal shock is irrelevant in both the *floating* and *pegged* regimes because it is fully funded. Since international financial shocks are absent in the *pegged* regime, nominal and real variables are driven solely by TFP shocks in that regime. In the *floating* regime, TFP shocks account for around half of inflation fluctuations, while international financial shocks account for the bulk of nominal and real exchange-rate fluctuations, consistent with Itskhoki and Mukhin (2025).

In the *fiscal* regime, unfunded fiscal shocks account for most of the variation in inflation and more than half of the variation in the nominal exchange rate. Since unfunded fiscal shocks affect the nominal but not the real exchange rate, the correlation between the two is substantially lower than in the *floating* regime.

Finally, Table C.1 shows that the two specifications of the simple model yield similar quantitative results, suggesting that the insights from the analytical framework carry over to the model with incomplete financial markets.

[9] In the next section, we assume that the central bank follows a Taylor rule that targets the nominal exchange rate, allowing it to exhibit positive but limited volatility.

Table 2: Model moments versus data moments

| | **Simple model** | **Quantitative model** | **Data** |
|---|---|---|---|
| *Floating regime (99Q1–19Q4)* | | | |
| $corr(\Delta e_t, \Delta q_t)$ | 1.00 | 1.00 | 1.00 |
| $corr(\Delta e_t, \pi_t)$ | 0.11 | 0.66 | 0.27 |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | 14 | 13 | 25 |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 0.23 | 0.23 | 0.28 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 3.11 | 2.67 | 6.74 |
| *Pegged regime (94Q3–98Q4)* | | | |
| $corr(\Delta e_t, \Delta q_t)$ | - | 0.73 | 0.93 |
| $corr(\Delta e_t, \pi_t)$ | - | 0.38 | 0.25 |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | - | 1.14 | 2.75 |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 1.10 | 0.64 | 0.41 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 1.09 | 0.81 | 1.10 |
| *Fiscal regime (80Q2–94Q2)* | | | |
| $corr(\Delta e_t, \Delta q_t)$ | 0.62 | 0.57 | 0.55 |
| $corr(\Delta e_t, \pi_t)$ | 0.86 | 0.88 | 0.67 |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | 1.27 | 1.20 | 1.16 |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 4.78 | 4.78 | 4.78 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 3.11 | 2.64 | 4.26 |

*Notes:* We do not report statistics involving the nominal exchange rate for the simple model under the *pegged* regime, as $\Delta e_t = 0$ makes them uninformative.

Table 3: Variance decomposition

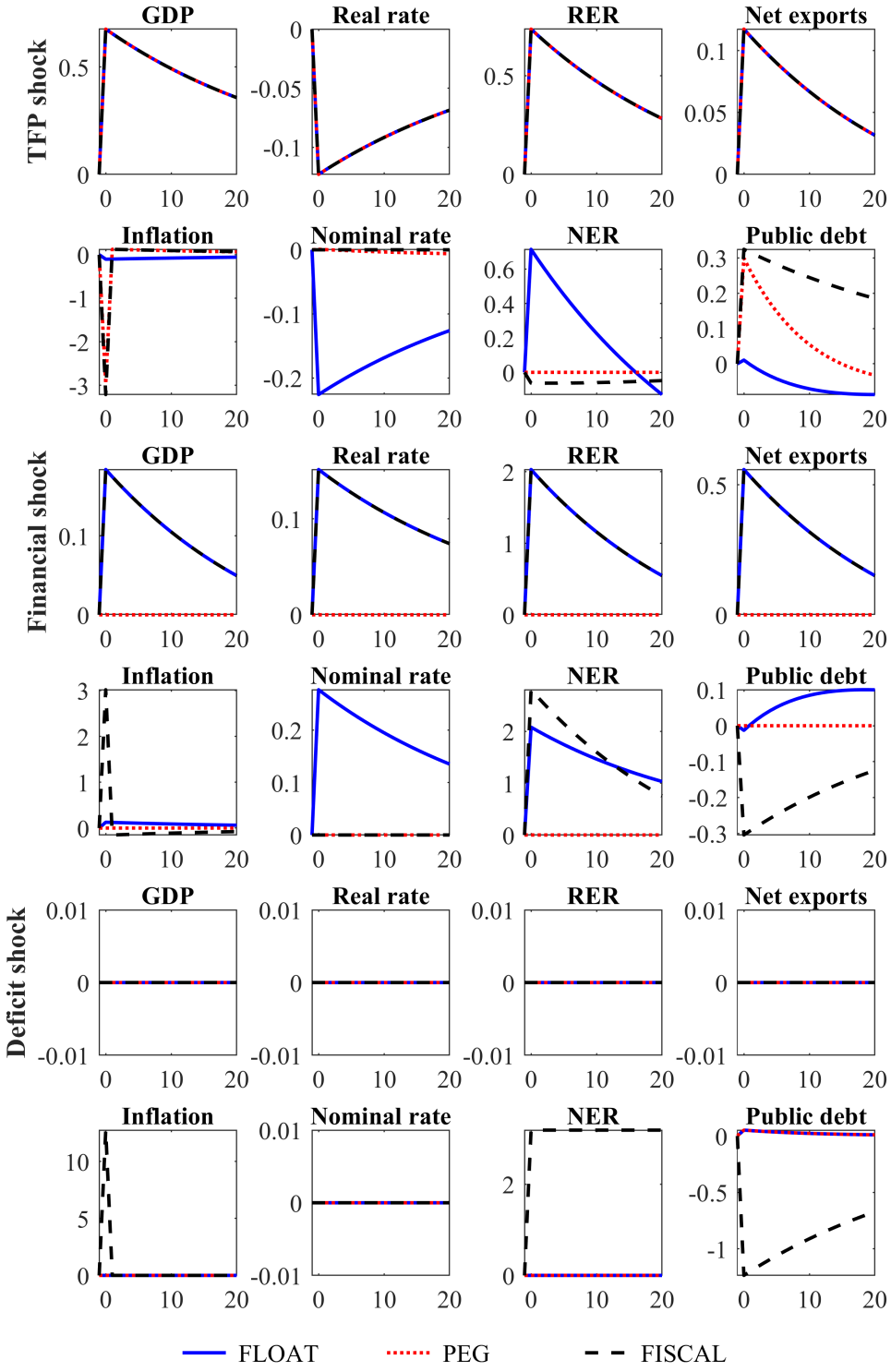
| *Floating* regime | $std(\pi_t)$ | $std(\Delta e_t)$ | $std(\Delta q_t)$ |
|---|---|---|---|
| TFP shock | 45 | 11 | 11 |
| Financial shock | 55 | 89 | 89 |
| Fiscal shock | 0 | 0 | 0 |
| *Fiscal* regime | $std(\pi_t)$ | $std(\Delta e_t)$ | $std(\Delta q_t)$ |
| TFP shock | 6 | 0 | 11 |
| Financial shock | 5 | 44 | 89 |
| Fiscal shock | 89 | 56 | 0 |

#### 2.9.3 Impulse response functions

We conclude the discussion of the simple model with impulse response functions to shocks of one standard deviation (Figure 3) . We report the response of GDP rather than output, while net exports are defined as the difference between GDP and consumption.

The responses are consistent with the analytical solution. The response of output to an international financial shock is theoretically ambiguous. Under our calibration, output and GDP increase because the expansion in foreign demand induced by the real depreciation more than offsets the decline in domestic consumption resulting from the higher real interest rate.

Figure 3: Impulse response functions in the simple model




*Notes:* GDP and exchange rates are reported in percentage deviations from steady state. Inflation and interest rates are in annualized deviations from steady state. Net exports are expressed as deviations from steady state divided by steady-state GDP. Public debt is expressed as deviations divided by annualized steady-state GDP. An increase in the real exchange rate (RER) or the nominal exchange rate (NER) denotes a depreciation.

# 3 Quantitative model

The goal of this section is to assess whether the results from the simple model carry over to a richer quantitative framework. To this end, we extend the model from the previous section along several dimensions, with a full description provided in the Appendix. First, we endogenize the foreign economy, yielding a two-country model, and denote by $n$ the home share of the world population. Second, we introduce capital and intermediate goods as additional production inputs. Third, we assume local currency pricing and adopt the Kimball aggregator to capture strategic complementarities in the price-setting decisions of firms.

Finally, we introduce money and a cash-in-advance (CIA) constraint on consumption $C_t$ and investment $Z_t$. This breaks the nominal dichotomy, so that real allocations also depend on nominal variables. We follow Schmitt-Grohé and Uribe (2007) and specify the CIA constraint as follows:

$$P_t(C_t + Z_t) \leq M_t, \tag{57}$$

where $M_t$ denotes nominal money.

The government budget constraint is now given by the following:

$$D_t + \frac{R_{t-1}}{\Pi_t} B_{Gt-1} = B_{Gt} + \frac{M_t}{P_t} - \frac{(M_{t-1}/P_{t-1})}{\Pi_t}. \tag{58}$$

Although this is not the conventional approach to breaking the nominal dichotomy—the literature typically relies on nominal price rigidities—it is particularly well suited to our setting, which focuses on periods of high inflation.[10] Moreover, our approach captures a common practice among many central banks during the final decades of the twentieth century: financing fiscal deficits through monetary expansion.

## 3.1 Calibration

We use a quarterly calibration, summarized in Table C.2. The population share $n$ is calibrated to $0.19$, which corresponds to the population share of Brazil in a world that consists only of Brazil and the G7 countries. We set openness to international trade $\gamma$ to $0.061$ to match the Brazilian import-to-GDP ratio of $12.2\%$. To obtain a symmetric steady state, we set the foreign openness to international trade at $\gamma^* = \gamma \frac{n}{1-n}$. We set the steady-state subsidy for firms at $\varsigma = \frac{1}{\theta}$, which simplifies the steady-state solution by compensating for the distortion of monopolistic competition. We set the investment adjustment-cost parameter $\kappa$ to match the Brazilian volatility of investment relative to output, which is $3.32$ for 1999-2019.

[10] As argued by Cavallo, Lippi, and Miyahara (2024), assuming price stickiness à la Calvo is unrealistic for environments with high and volatile inflation.

Steady-state TFP $\bar{a}$ is calibrated to normalize total output to one. We set the standard deviation of the international financial shock $\sigma_\psi$ to match the correlation between the Brazilian-G7 consumption growth differential and the Brazilian real exchange-rate growth for 1999-2019 ($-0.22$). We set the cross-country TFP correlation $\rho_{a,a^*}$ to match the correlation between Brazilian and G7 output, which is $0.37$ since 1999. We set the standard deviation of the fiscal shock to match the standard deviation of inflation relative to output during 1980-1994 ($4.78$). The policy parameters are discussed in the next paragraph.

### 3.1.1 Policy parameters

We analyze the same three regimes as in the previous section, summarized in Table C.3. We consider a more general framework, where the rules have inertia (see parameter $\rho_m = 0.95$) and, in the *fiscal* regime, the central bank can respond to inflation with $\phi_\pi^{fiscal} < 1$.

In the *floating* regime, the central bank adopts a flexible exchange rate and responds only to inflation, with $\phi_\pi^{floating} = 2.15$ and an annual inflation target of $3\%$. In the *pegged* regime, we calibrate the Taylor-rule coefficients on inflation and the nominal exchange rate to match the ratios of the standard deviations of $\pi_t$ and $\Delta e_t$ in the *floating* regime relative to the *pegged* regime, which are $0.43$ and $3.87$ in the data, respectively. This yields $\phi_\pi^{pegged} = 0.87$ and $\phi_e = 0.36$. Since the peg is not perfect, we also reduce $\Gamma$ (and hence $\iota$) in equation (54) by a factor of $(3.87)^2$, reflecting the lower variance of the nominal exchange rate during July 1994–December 1998 relative to January 1999–December 2019. Under a perfectly pegged exchange rate, $\Gamma$ would be zero.

In the *fiscal* regime, we set $\phi_\pi^{fiscal} = 0.8$, so that the Taylor principle is not satisfied. We set the steady-state public debt stock at $40.25\%$ of GDP ($\bar{B}_G = 0.80$). We also set the steady-state public debt in the *floating* and *pegged* regimes to the same level as in the *fiscal* regime, which is calibrated using data from the 1980–1994 period; this assumption is innocuous, as the steady-state level of public debt is irrelevant under these regimes and corresponds to a fiscal surplus of $0.4\%$ of GDP.

However, a steady-state fiscal surplus in the *fiscal* regime would be highly counterfactual, as this period was characterized by large primary deficits. We therefore calibrate a positive steady-state primary deficit of $3.37\%$ of GDP ($\bar{D} = 0.0169$), which matches the Brazilian average over 1981–1994. Given $\bar{D}$ and $\bar{B}_G$, the government budget constraint implies that steady-state inflation differs from that in the *pegged* and *floating* regimes, with an implied quarterly inflation rate of $9\%$.[11] In Subsection 3.3, we show that imposing the same steady state for nominal variables across regimes does not affect our results.

Lastly, in the foreign country, the central bank follows a floating exchange-rate regime with $\phi_\pi^{floating} = 2.15$ and $\Pi^* = 1.005$, which implies an annual inflation target of $2\%$.

[11] In a model with a CIA constraint, the steady-state inflation rate affects real variables. To ensure that real variables share the same steady state across regimes and between the home and foreign economies, we introduce steady-state subsidies to labor and capital. This assumption substantially simplifies the steady-state algebra.

## 3.2 Results

The second column of Table 2 reports the key moments generated by the quantitative model. The main results obtained with the simple model continue to hold. In particular, the quantitative model replicates the decline in the correlation between the nominal and the real exchange rates when moving from the *floating* to the *fiscal* regime. Moreover, it also captures the high correlation between the nominal and real exchange rates in the *pegged* regime. Under the pegged regime in the simple model, on the contrary, the volatility of the nominal exchange rate was assumed to be zero.

The variance decomposition is also broadly consistent with that of the simple model. The main differences are that foreign TFP shocks are now included and that international financial shocks are present even in the *pegged* regime, albeit with relatively low volatility. We continue to find that the nominal exchange rate is driven primarily by financial shocks under the *floating* regime and by deficit shocks under the *fiscal* regime.

The impulse response functions also reflect our analytical results (Figures 4 and 5), with some differences. First, we introduce a new shock: an increase in foreign TFP. This raises demand for the domestic good and generates a modest expansion in the home economy. The shock induces both a real and nominal appreciation of the home currency in all regimes, as the domestic good becomes relatively scarcer. Inflation increases under the *floating* regime because the real interest rate rises (see equation 30). Under the *pegged* regime, inflation declines following the real appreciation (see equation 36). Under the *fiscal* regime, inflation also declines, as the increase in consumption raises the demand for real money balances, allowing the government to rely less on inflationary finance to fund the deficit.

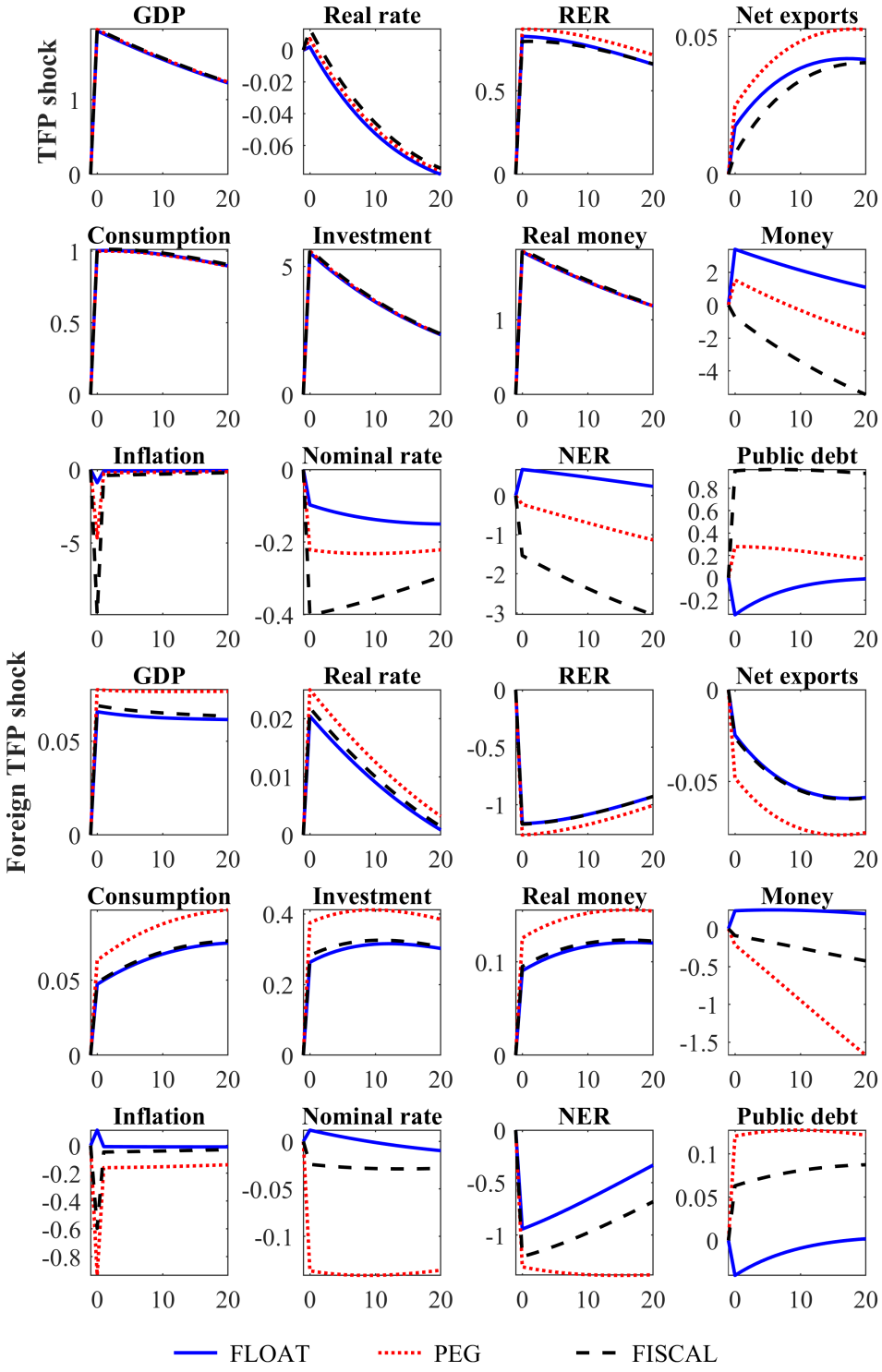
Second, the CIA constraint breaks the monetary dichotomy, implying that real variables respond differently between the three regimes. Remarkably, under the *fiscal* regime, an unfunded fiscal shock raises inflation, thereby reducing economic activity by lowering real money balances.

Third, because the exchange-rate peg is no longer perfect, international financial shocks operate even under the *pegged* regime, although their volatility remains low.

Table 4: Variance decomposition

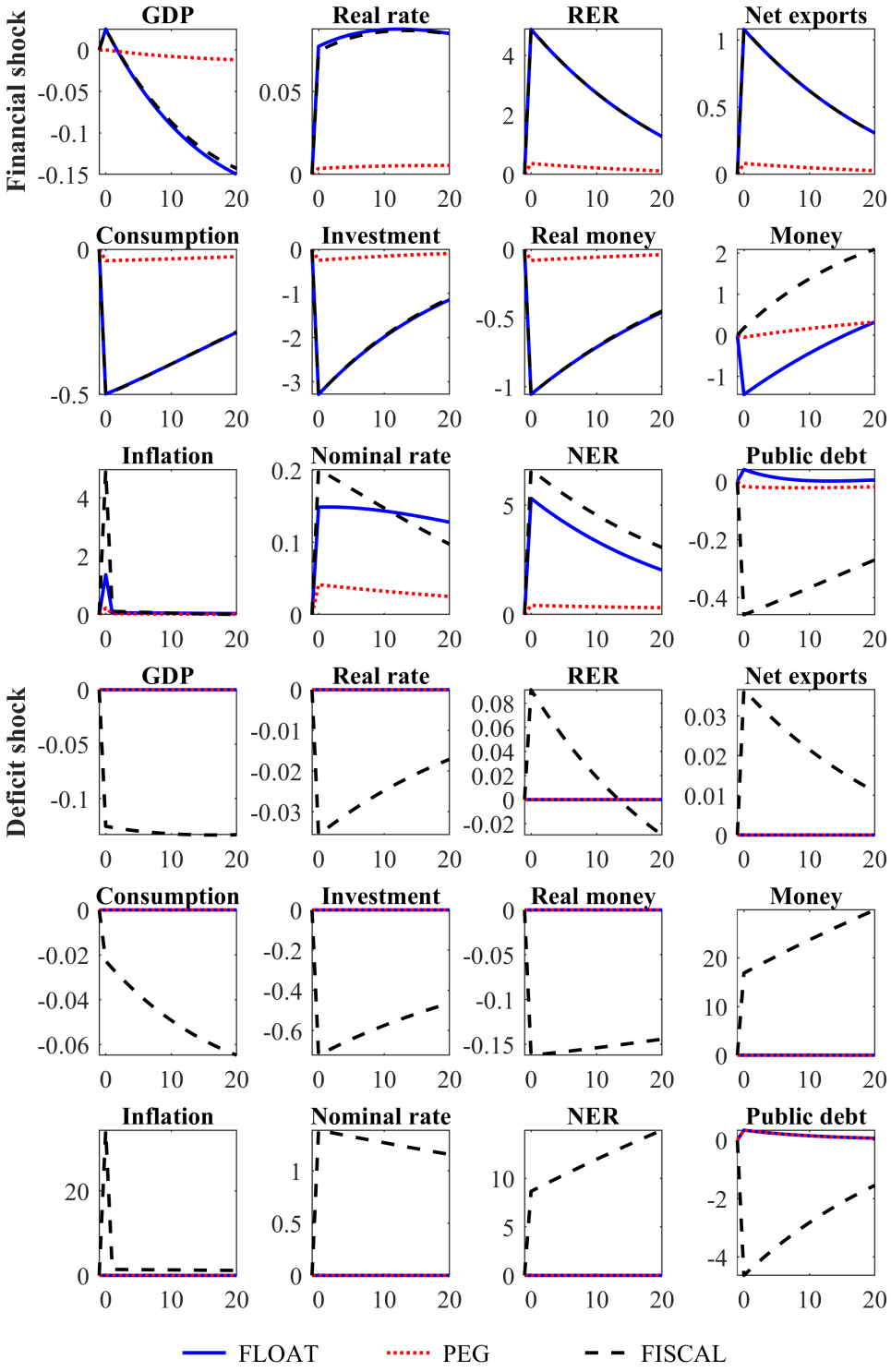
| *Floating* Regime | $std(\pi_t)$ | $std(\Delta e_t)$ | $std(\Delta q_t)$ |
|---|---|---|---|
| Home TFP shock | 35 | 2 | 3 |
| Foreign TFP shock | 1 | 3 | 5 |
| Financial shock | 64 | 95 | 92 |
| Fiscal shock | 0 | 0 | 0 |
| *Pegged* Regime | $std(\pi_t)$ | $std(\Delta e_t)$ | $std(\Delta q_t)$ |
| Home TFP shock | 92 | 7 | 31 |
| Foreign TFP shock | 8 | 84 | 64 |
| Financial shock | 0 | 9 | 5 |
| Fiscal shock | 0 | 0 | 0 |
| *Fiscal* Regime | $std(\pi_t)$ | $std(\Delta e_t)$ | $std(\Delta q_t)$ |
| Home TFP shock | 5 | 1 | 3 |
| Foreign TFP shock | 0 | 1 | 5 |
| Financial shock | 1 | 32 | 92 |
| Fiscal shock | 94 | 66 | 0 |

Figure 4: Impulse response functions in the quantitative model (1)




*Notes:* GDP, exchange rates, real and nominal money, consumption, and investment are reported in percentage deviations from steady state. Inflation and interest rates are in annualized deviations from steady state. Net exports are expressed as deviations from steady state divided by steady-state GDP. Public debt is expressed as deviations divided by annualized steady-state GDP. An increase in the real exchange rate (RER) or the nominal exchange rate (NER) denotes a depreciation.

Figure 5: Impulse response functions in the quantitative model (2)




*Notes:* GDP, exchange rates, real and nominal money, consumption, and investment are reported in percentage deviations from steady state. Inflation and interest rates are in annualized deviations from steady state. Net exports are expressed as deviations from steady state divided by steady-state GDP. Public debt is expressed as deviations divided by annualized steady-state GDP. An increase in the real exchange rate (RER) or the nominal exchange rate (NER) denotes a depreciation.

## 3.3 Sensitivity analysis

We conduct three additional exercises for the *fiscal* regime to assess the robustness of our results to alternative modeling assumptions. First, we assume that fiscal shocks are financed through changes in the money supply. Second, we modify the calibration of the standard deviation of the fiscal shock. Third, we impose the same steady state for the *fiscal* regime as for the other two regimes. The resulting statistics are reported in Table C.4. The variance decompositions and impulse response functions remain qualitatively unchanged and are available upon request.

### 3.3.1 Money supply

Following Sargent (2013) and Kehoe and Nicolini (2022), we introduce a binding government debt constraint:

$$B_{Gt} = \bar{B}_G. \tag{59}$$

The debt constraint captures the idea that, once public debt becomes sufficiently high, the government can no longer finance additional deficits through further debt accumulation. Under this restriction, money creation and inflation adjust jointly to satisfy the government budget constraint. Equation (59) is similar to a monetary policy rule, which requires monetary financing whenever a fiscal shock occurs; accordingly, it replaces the Taylor rule.

To preserve the target inflation-to-output volatility ratio, we recalibrate the standard deviation of fiscal shocks. The main results remain qualitatively unchanged (Table C.4, second column). The impulse response to a fiscal shock is also qualitatively similar to the baseline. The main difference is that inflation responds less on impact but remains elevated for longer, while the decline in output is more pronounced.

### 3.3.2 Deficit volatility

In the baseline model, the standard deviation of fiscal shocks is calibrated to match the target inflation-to-output volatility ratio. As an alternative calibration, we instead set the standard deviation of the fiscal shock to match the standard deviation of the Brazilian deficit during the period 1980–1994. Since quarterly data on the Brazilian deficit are unavailable for this period, we infer the standard deviation of the quarterly deficit $\sigma_d$ from the annual deficit series reported in Kehoe and Nicolini (2022).[12] Under this calibration, the inflation-to-output volatility ratio declines from $4.78$ to $2.17$. The two correlations involving the nominal exchange rate move in opposite directions.

[12] We use the standard deviation of the annual series, $\sigma_{Ad}$, together with the quarterly autocorrelation $\rho_{Qd} = \rho_{Ad}^{0.25}$ obtained from the annual autocorrelation $\rho_{Ad}$, to infer the standard deviation of the quarterly deficit, $\sigma_d$, using the following formula:

$$\sigma_d^2 = \frac{\sigma_{Ad}^2 \left(1 - \rho_{Qd}^2\right)}{4 + 2\left(3\rho_{Qd} + 2\rho_{Qd}^2 + \rho_{Qd}^3\right)},$$

which yields $\sigma_d = 0.59$.

The correlation between the nominal and the real exchange rate rises to $0.83$: although higher than in the baseline, it remains well below the near-unit value of the *floating* regime, so the breakdown of the comovement under fiscal dominance is preserved. At the same time, the correlation between the nominal exchange rate and inflation falls to $0.70$, much closer to the data (Table C.4, third column).

#### 3.3.3 Same steady state

In the baseline model, the *fiscal* regime has a different steady state for nominal variables. We now impose the same steady state across regimes by assuming a steady-state fiscal surplus in the *fiscal* regime as well. The main results remain qualitatively unchanged (Table C.4, fourth column).

## 4 Conclusions

With public debt in advanced economies at historically high levels and growing concerns about fiscal sustainability, there is an increasing risk that fiscal considerations will play a more prominent role in determining inflation in the near future. When the fiscal authority sets the primary balance independently of the outstanding stock of public debt—that is, under fiscal dominance—the price level has to adjust to ensure that the intertemporal budget constraint of the government is satisfied. Nevertheless, considerably less attention has been paid to the implications of fiscal dominance for the dynamics of nominal and real exchange rates. This paper fills this gap by developing a theoretical framework to study the determination of exchange rates, as well as their relationship to macroeconomic aggregates, under alternative monetary–fiscal policy interactions.

When fiscal policy is passive and fiscal shocks are fully funded, monetary policy anchors inflation, and exchange rates remain unaffected by fiscal policy decisions. By contrast, when the fiscal authority does not adjust future primary surpluses in response to an increase in public debt, the price level is determined by the intertemporal budget constraint of the government. In this case, an unfunded fiscal shock generates inflation because the real value of public debt can no longer be stabilized through future fiscal adjustments. The resulting increase in the inflation rate is accompanied by a proportional depreciation of the nominal exchange rate, thereby breaking the comovement between nominal and real exchange rates that typically arises in response to productivity disturbances and international financial flows.

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# Appendix

## A Simple Model

### A.1 Steady state

The steady state is given by:

$$
\begin{aligned}
Y_H = C = Q = P_H = 1 \\
RR = \frac{1}{\beta} \\
B = 0 \\
\Pi = \Delta E = \bar{\Pi} \\
R = \frac{\bar{\Pi}}{\beta} \\
B_G = \bar{B}_G.
\end{aligned}
$$

### A.2 Real block of the linearized simple model

Lower-case variables denote percentage deviations from the steady state, except for $b_{Gt}$, $b_t$, and $d_t$, which denote absolute deviations from the steady state. The real block is given by equations (22)-(25). Using $q_t = \sigma\left(c_t + \psi_t\right)$, we get:

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma} r r_t \tag{A.1}$$

$$y_{Ht} = \frac{\omega}{1-\gamma} c_t + \theta\gamma\sigma \frac{2-\gamma}{1-\gamma} \psi_t \tag{A.2}$$

$$c_t = -\gamma\sigma \frac{\nu + \theta\left(2-\gamma\right)}{\omega + \sigma\nu} \psi_t + \frac{\left(1+\nu\right)\left(1-\gamma\right)}{\omega + \sigma\nu} a_t, \tag{A.3}$$

where $\omega \equiv \left(1-\gamma\right)^2 + \theta\gamma\sigma\left(2-\gamma\right)$. Combining the last two equations yields the following solution for consumption:

$$c_t = -\gamma\sigma \frac{\nu + \theta\left(2-\gamma\right)}{\omega + \sigma\nu} \psi_t + \frac{\left(1+\nu\right)\left(1-\gamma\right)}{\omega + \sigma\nu} a_t, \tag{A.4}$$

and output:

$$y_{Ht} = \gamma\sigma\nu \frac{\theta\sigma\left(2-\gamma\right) - \left(1-\gamma\right)}{\omega + \sigma\nu} \psi_t + \omega \frac{1+\nu}{\omega + \sigma\nu} a_t. \tag{A.5}$$

Using the Euler equation and the Backus-Smith condition, we get:

$$rr_t = -\Psi_a a_t + \Psi_\psi \psi_t \tag{A.6}$$

$$q_t = \Xi_\psi \psi_t + \Xi_a a_t, \tag{A.7}$$

where $\Psi_a \equiv \frac{\sigma(1-\rho_a)(1+\nu)(1-\gamma)}{\omega+\sigma\nu}$, $\Psi_\psi \equiv \frac{\gamma(1-\rho_\psi)\sigma^2[\nu+\theta(2-\gamma)]}{\omega+\sigma\nu}$, $\Xi_\psi \equiv \sigma\,(1-\gamma)\,\frac{1-\gamma+\sigma\nu}{\omega+\sigma\nu}$, and $\Xi_a \equiv \sigma\,\frac{(1+\nu)(1-\gamma)}{\omega+\sigma\nu}$.

## A.3 Nominal block of the linearized simple model: *floating* regime

Combining the Taylor rule with the Fisher equation:

$$\pi_t = \frac{1}{\phi_\pi}\left(rr_t + \mathbb{E}_t\pi_{t+1}\right) \tag{A.8}$$

$$\pi_t = \frac{1}{\phi_\pi}\sum_{j=0}^{\infty}\left(\frac{1}{\phi_\pi}\right)^j rr_{t+j} \tag{A.9}$$

$$\pi_t = \frac{1}{\phi_\pi}\sum_{j=0}^{\infty}\left(\frac{1}{\phi_\pi}\right)^j\left(\Psi_\psi\rho_\psi^j\psi_t - \Psi_a\rho_a^j a_t\right) \tag{A.10}$$

$$\pi_t = \frac{\Psi_\psi}{\phi_\pi}\psi_t\sum_{j=0}^{\infty}\left(\frac{\rho_\psi}{\phi_\pi}\right)^j - \frac{\Psi_a}{\phi_\pi}a_t\sum_{j=0}^{\infty}\left(\frac{\rho_a}{\phi_\pi}\right)^j \tag{A.11}$$

$$\pi_t = \frac{\Psi_\psi}{\phi_\pi-\rho_\psi}\psi_t - \frac{\Psi_a}{\phi_\pi-\rho_a}a_t, \tag{A.12}$$

since inflation converges to zero in the long run. Given inflation, we can easily find the nominal rate:

$$r_t = \phi_\pi\pi_t, \tag{A.13}$$

and the nominal exchange rate, using the solution for the real exchange rate:

$$\Delta e_t = \pi_t + q_t - q_{t-1} \tag{A.14}$$

$$\Delta e_t = \frac{\Psi_\psi}{\phi_\pi-\rho_\psi}\psi_t - \frac{\Psi_a}{\phi_\pi-\rho_a}a_t + \Xi_\psi\left(\psi_t-\psi_{t-1}\right) + \Xi_a\left(a_t-a_{t-1}\right). \tag{A.15}$$

$$\Delta e_t = \frac{\phi_\pi-1}{\phi_\pi-\rho_a}\Xi_a a_t - \Xi_a a_{t-1} + \frac{\Psi_\psi}{\phi_\pi-\rho_\psi}\psi_t + \Xi_\psi\left(\psi_t-\psi_{t-1}\right). \tag{A.16}$$

## A.4 Nominal block of the linearized simple model: *pegged* regime

The exchange rate is determined by the monetary rule $\Delta e_t = 0$, which implies that inflation depends only on $q_t$:

$$\pi_t = -\left(q_t - q_{t-1}\right) \tag{A.17}$$

$$\pi_t = -\Xi_a \left(a_t - a_{t-1}\right). \tag{A.18}$$

The nominal interest rate is given by the Fisher equation:

$$r_t = -\Psi_a a_t - \Xi_a \mathbb{E}_t \left(a_{t+1} - a_t\right) \tag{A.19}$$

$$r_t = -\Psi_a a_t + \left(1 - \rho_a\right) \Xi_a a_t \tag{A.20}$$

$$r_t = \left[-\Psi_a + \left(1 - \rho_a\right) \Xi_a\right] a_t = 0. \tag{A.21}$$

## A.5 Nominal block of the linearized simple model: *fiscal* regime

The deficit is determined by its purely exogenous process:

$$d_t = \rho_d d_{t-1} + v_t^d. \tag{A.22}$$

The nominal rate is constant ($r_t = 0$). Inflation and public debt are jointly determined by the budget constraint of the government and the Fisher equation:

$$\pi_t = \frac{1}{B_G}\left[b_{Gt-1} + \beta\left(d_t - b_{Gt}\right)\right] \tag{A.23}$$

$$\mathbb{E}_t \pi_{t+1} = -r r_t. \tag{A.24}$$

Iterating the first equation forward:

$$\mathbb{E}_t \pi_{t+1} = \frac{1}{B_G}\mathbb{E}_t\left[b_{Gt} + \beta\left(d_{t+1} - b_{Gt+1}\right)\right] \tag{A.25}$$

$$\Psi_a a_t - \Psi_\psi \psi_t = \frac{1}{B_G}\mathbb{E}_t\left[b_{Gt} + \beta\left(\rho_d d_t - b_{Gt+1}\right)\right] \tag{A.26}$$

$$b_{Gt} = -B_G \Psi_\psi \psi_t + B_G \Psi_a a_t - \beta \rho_d d_t + \beta b_{Gt+1} \tag{A.27}$$

$$b_{Gt} = -B_G \Psi_\psi \sum_{j=0}^{\infty} \left(\beta \rho_\psi\right)^j \psi_t + B_G \Psi_a \sum_{j=0}^{\infty} \left(\beta \rho_a\right)^j a_t - \beta \rho_d \sum_{j=0}^{\infty} \left(\beta \rho_d\right)^j d_t \tag{A.28}$$

$$b_{Gt} = -\frac{B_G \Psi_\psi}{1 - \beta \rho_\psi}\psi_t + \frac{B_G \Psi_a}{1 - \beta \rho_a} a_t - \frac{\beta \rho_d}{1 - \beta \rho_d} d_t. \tag{A.29}$$

Thus:

$$\pi_t = -\frac{\beta}{B_G} b_{Gt} + \frac{1}{B_G} b_{Gt-1} + \frac{\beta}{B_G} d_t \tag{A.30}$$

$$\pi_t = -\frac{\beta}{B_G}\left[-\frac{B_G\Psi_\psi}{1-\beta\rho_\psi}\psi_t + \frac{B_G\Psi_a}{1-\beta\rho_a}a_t - \frac{\beta\rho_d}{1-\beta\rho_d}d_t\right] + \frac{1}{B_G}\left[-\frac{B_G\Psi_\psi}{1-\beta\rho_\psi}\psi_{t-1} + \frac{B_G\Psi_a}{1-\beta\rho_a}a_{t-1} - \frac{\beta\rho_d}{1-\beta\rho_d}d_{t-1}\right] + \frac{\beta}{B_G}d_t \tag{A.31}$$

$$\pi_t = \frac{\Psi_\psi}{1-\beta\rho_\psi}\left(\beta\psi_t - \psi_{t-1}\right) - \frac{\Psi_a}{1-\beta\rho_a}\left(\beta a_t - a_{t-1}\right) + \frac{\beta}{B_G}\left(1 + \frac{\beta\rho_d}{1-\beta\rho_d}\right)d_t - \frac{1}{B_G}\frac{\beta\rho_d}{1-\beta\rho_d}d_{t-1} \tag{A.32}$$

$$\pi_t = \frac{\Psi_\psi}{1-\beta\rho_\psi}\left(\beta\psi_t - \psi_{t-1}\right) - \frac{\Psi_a}{1-\beta\rho_a}\left(\beta a_t - a_{t-1}\right) + \frac{\beta}{B_G\left(1-\beta\rho_d\right)}\left(d_t - \rho_d d_{t-1}\right). \tag{A.33}$$

Finally, the nominal exchange rate is determined by:

$$\Delta e_t = \pi_t + q_t - q_{t-1} \tag{A.34}$$

$$\Delta e_t = \frac{\beta}{B_G\left(1-\beta\rho_d\right)}\left(d_t - \rho_d d_{t-1}\right) + \frac{\Psi_\psi}{1-\beta\rho_\psi}\left(\beta\psi_t - \psi_{t-1}\right) - \frac{\Psi_a}{1-\beta\rho_a}\left(\beta a_t - a_{t-1}\right) + \Xi_\psi\left(\psi_t - \psi_{t-1}\right) + \Xi_a\left(a_t - a_{t-1}\right). \tag{A.35}$$

# B Quantitative Model

We describe the main differences relative to the simple model, focusing on the home economy. The foreign economy is symmetric. Again, variables denoted by a superscript'*' refer to the foreign economy.

## B.1 Households

### B.1.1 Intertemporal problem

Compared to the simple model, the budget constraint reads:

$$\begin{aligned} C_t + K_t + B_t + B_{Gt} + \frac{M_t}{P_t} =& \frac{W_t}{\zeta_L}L_t + \left[\left(1+\zeta_K\right)R_t^K + \left(1-\delta\right)\right]K_{t-1} + \frac{R_{t-1}\left(B_{t-1}+B_{Gt-1}\right)}{\Pi_t} + \\ &+ \frac{\left(M_{t-1}/P_{t-1}\right)}{\Pi_t} + Prof_t - T_t - \frac{\kappa}{2}\frac{\left(K_t - K_{t-1}\right)^2}{K_{t-1}}, \end{aligned} \tag{B.1}$$

where $M_t$ denotes nominal money; $R_t^K$ and $\delta$ are the capital rental rate and the depreciation rate, respectively; and $\zeta_K$ and $\zeta_L$ are wedges needed to make the steady state identical in the two countries under the *fiscal* regime; $K_t$ denotes the capital stock, which is subject to quadratic adjustment costs measured by $\kappa > 0$.

$K_t$ obeys the following law of motion:

$$K_t = (1-\delta)K_{t-1} + \left[ Z_t - \frac{\kappa}{2}\frac{(K_t - K_{t-1})^2}{K_{t-1}} \right]. \tag{B.2}$$

Here, $Z_t$ is investment.

We also introduce a cash-in-advance (CIA) constraint, following Schmitt-Grohé and Uribe (2007):

$$P_t(C_t + Z_t) \leq M_t. \tag{B.3}$$

This problem yields a Euler equation for public and private bonds:

$$1 = \beta \mathbb{E}_t \left( \frac{\lambda^c_{t+1}}{\lambda^c_t} \frac{R_t}{\Pi_{t+1}} \right), \tag{B.4}$$

where $\lambda^c_t$ is the Lagrange multiplier on the budget constraint:

$$\lambda^c_t = C_t^{-\sigma} - \lambda^m_t, \tag{B.5}$$

and $\lambda^m_t$ is the Lagrange multiplier on the CIA constraint.

This problem also yields a Euler equation for capital:

$$\begin{aligned} 1 = \beta \mathbb{E}_t \frac{1}{(\lambda^c_t + \lambda^m_t)} \left[ \lambda^c_{t+1} (1+\zeta_K) R^K_{t+1} + (1-\delta)\left(\lambda^c_{t+1} + \lambda^m_{t+1}\right) \right] - \kappa \frac{(K_t - K_{t-1})}{K_{t-1}} + \\ + \beta \mathbb{E}_t \left[ \frac{\lambda^c_{t+1} + \lambda^m_{t+1}}{\lambda^c_t + \lambda^m_t} \right] \left[ \frac{\kappa (K_{t+1} - K_t)}{K_t} + \frac{\kappa}{2} \frac{(K_{t+1} - K_t)^2}{K_t^2} \right]; \end{aligned} \tag{B.6}$$

a labor supply condition:

$$\zeta_L L_t^{\frac{1}{\nu}} = W_t \lambda^c_t; \tag{B.7}$$

and a Euler equation for money:

$$\lambda^c_t = \lambda^m_t + \beta \mathbb{E}_t \frac{\lambda^c_{t+1}}{\Pi_{t+1}}. \tag{B.8}$$

### B.1.2 Intratemporal problem

The consumption bundle $C_t$ consists of a continuum of differentiated domestic goods $C_{Ht}(i)$ and a continuum of differentiated foreign goods $C_{Ft}(i)$, and is implicitly defined by the Kimball (1995) aggregator as follows:

$$\int_0^1 \left[ (1-\gamma) g\left( \frac{C_{Ht}(i)}{(1-\gamma) C_t} \right) + \gamma g\left( \frac{C_{Ft}(i)}{\gamma C_t} \right) \right] di = 1. \tag{B.9}$$

Here, $\gamma > 0$ captures home bias in international trade, and the function $g(\cdot)$ is increasing and concave, with $-g''(1) \in (0,1)$ and $g(1) = g'(1) = 1$. The representative household minimizes total expenditure:

$$\min \int_0^1 \left[ P_{Ht}^N(i)\, C_{Ht}(i) + P_{Ft}^N(i)\, C_{Ft}(i) \right] di, \tag{B.10}$$

given the consumption bundle in equation (B.9), where $P_{Ht}^N(i)$ and $P_{Ft}^N(i)$ are the nominal prices of the individual home and foreign varieties. The minimization problem results in the following demand schedules:

$$C_{Ht}(i) = (1-\gamma) h\left(\frac{P_{Ht}^N(i)}{\mathcal{P}_t}\right) C_t \qquad \text{and} \qquad C_{Ft}(i) = \gamma h\left(\frac{P_{Ft}^N(i)}{\mathcal{P}_t}\right) C_t,$$

where $h(\cdot) = g'^{-1}(\cdot)$, and $\theta_t \equiv -\frac{\partial \log h(x)}{\partial \log x}$ is the price elasticity of the demand schedule, which is not constant under the Kimball aggregator; we denote by $\theta \equiv \theta(1) = -h'(1)$ its steady-state value.

The consumer price index $P_t$ and the auxiliary variable $\mathcal{P}_t$ are implicitly defined by the consumption-expenditure equation and by the Kimball (1995) aggregator, after substituting the demand schedules:

$$P_t = \int_0^1 \left[ (1-\gamma) P_{Ht}^N(i)\, h\left(\frac{P_{Ht}^N(i)}{\mathcal{P}_t}\right) + \gamma P_{Ft}^N(i)\, h\left(\frac{P_{Ft}^N(i)}{\mathcal{P}_t}\right) \right] di, \tag{B.11}$$

$$\int_0^1 \left\{ (1-\gamma) g\left[ h\left(\frac{P_{Ht}^N(i)}{\mathcal{P}_t}\right)\right] + \gamma g\left[ h\left(\frac{P_{Ft}^N(i)}{\mathcal{P}_t}\right)\right] \right\} di = 1. \tag{B.12}$$

## B.2 Firms

Each home firm $i$ produces the differentiated domestic good $i$, operating in monopolistic competition, and using the following production function:

$$Y_{Ht}(i) + Y_{Et}(i) = \left( e^{a_t} K_{t-1}(i)^{\vartheta} L_t(i)^{1-\vartheta} \right)^{1-\phi} X_t^{\phi}(i).$$

Here, $Y_{Ht}(i)$ is domestic good $i$ sold to home households, $Y_{Et}(i)$ is domestic good $i$ exported to foreign households; $\vartheta \in (0,1)$ measures the capital share in value added; $\phi \in (0,1)$ measures the elasticity of the output with respect to the intermediate input; $a_t$ and its foreign counterpart $a_t^*$ are the logarithms of total factor productivity.

The logarithms of total factor productivity follow a multivariate AR(1) process:

$$a_t = (1-\rho_a)\,\bar{a} + \rho_a a_{t-1} + \epsilon_t^a \tag{B.13}$$

$$a_t^* = (1-\rho_a)\,\bar{a} + \rho_a a_{t-1}^* + \epsilon_t^{a*} \tag{B.14}$$

$$\begin{bmatrix} \varepsilon_t^a \\ \varepsilon_t^{a*} \end{bmatrix} \sim \mathcal{N}\left( \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \rho_{a,a^*}\sigma_a^2 \\ \rho_{a,a^*}\sigma_a^2 & \sigma_a^2 \end{bmatrix} \right) \tag{B.15}$$

where $\epsilon_t^a$ is a TFP shock, $\bar{a}$ is steady-state TFP, $\rho_a \in [0,1)$ is the autoregressive parameter, $\sigma_a \geq 0$ is the standard deviation of TFP shocks, and $\rho_{a,a^*} \in [-1,1]$ is the cross-country correlation.

The intermediate input $X_t$ is a bundle of domestic and foreign varieties, as for final consumption $C_t$ and gross investment $Z_t$, aggregated according to an analogous Kimball (1995) aggregator and demanded according to analogous demand schedules. Given that all firms $i$ are identical, they make the same choices, and from now on we omit the $i$ index. Firms have market power and can set a different price in each market. The price is denominated in the currency of the buyer. The first-order conditions of the profit maximization yield the following three input demands:

$$(1-\varsigma)\,W_t L_t = (1-\phi)\,(1-\vartheta)\,MC_t\,(Y_{Ht} + Y_{Et}) \tag{B.16}$$

$$(1-\varsigma)\,R_t^K K_{t-1} = (1-\phi)\,\vartheta MC_t\,(Y_{Ht} + Y_{Et}) \tag{B.17}$$

$$(1-\varsigma)\,X_t = \phi MC_t\,(Y_{Ht} + Y_{Et})\,, \tag{B.18}$$

Here, $MC_t$ denotes the real marginal cost, and $\varsigma$ is a subsidy to firms' costs, introduced for the sole purpose of easing the computation of the steady state. Optimal pricing in the home market reads:

$$P_{Ht} = \mu\left(\frac{P_{Ht}^N}{\mathcal{P}_t}\right) MC_t, \tag{B.19}$$

where $\mu(x) \equiv \frac{\theta_t(x)}{\theta_t(x)-1}$ is the markup function and $\theta_t(x)$ is the price elasticity of the demand schedule, defined in the previous subsection. After a first-order Taylor approximation, we get:

$$p_{Ht} = (1-\alpha)\,mc_t. \tag{B.20}$$

Here,

$$\alpha \equiv -\frac{\frac{\partial \log \mu(x)}{\partial \log x}}{1 - \frac{\partial \log \mu(x)}{\partial \log x}} \tag{B.21}$$

captures the elasticity of strategic complementarities between firms in price setting, and lower-case letters are log-deviations from the steady state.

A special case of the Kimball aggregator is the standard CES bundle, where $\theta$, and thus $\mu$, are constant, and $\alpha = 0$. Finally, optimal pricing in the foreign market reads:

$$P_{Ht}^{*} = \mu \left( \frac{P_{Ht}^{N*}}{\mathcal{P}_{t}^{*}} \right) \frac{MC_t}{Q_t}, \tag{B.22}$$

where $E_t$ is the nominal exchange rate, defined as in the simple model.

## B.3 Incomplete financial markets

International financial markets follow the same formulation as in the simple model.

## B.4 Policy

The government budget constraint reads:

$$D_t + \frac{R_{t-1}}{\Pi_t} B_{Gt-1} = B_{Gt} + \frac{M_t}{P_t} - \frac{(M_{t-1}/P_{t-1})}{\Pi_t}, \tag{B.23}$$

where the right-hand side now also features money creation.

Monetary policy follows a Taylor rule with interest-rate smoothing, whose active terms depend on the regime, while fiscal policy sets the primary deficit:

$$R_t = R^{\rho_m} \left[ \left( \frac{\Pi_t}{\bar{\Pi}} \right)^{floating \cdot \phi_\pi^{floating} + pegged \cdot \phi_\pi^{pegged} + fiscal \cdot \phi_\pi^{fiscal}} \left( \frac{\Delta E_t}{\Delta \bar{E}} \right)^{pegged \cdot \phi_e} \right]^{1-\rho_m} \tag{B.24}$$

$$\frac{D_t - \bar{D}}{GDP} = \left[ - \left( floating + pegged \right) \phi_b \frac{B_{Gt-1} - \bar{B_G}}{GDP} + fiscal \cdot \rho_d \frac{D_{t-1} - \bar{D}}{GDP} \right] + v_t^d. \tag{B.25}$$

Here, $\rho_m \in [0, 1)$ governs the smoothing of the interest rate.

The three regimes differ in the active terms of the monetary rule and in the stance of fiscal policy. Under the *floating* regime, the central bank follows an active Taylor rule with $\phi_\pi^{floating} > 1$, and fiscal policy is passive with $-1 + \beta^{-1} < \phi_b < 1 + \beta^{-1}$. Under the *pegged* regime, the central bank responds to both inflation and the nominal exchange rate, with $\phi_\pi^{pegged}$ and $\phi_e$; since the peg is imperfect, the nominal exchange rate retains a positive but limited volatility. Under the *fiscal* regime, monetary policy is (weakly) passive, with $\phi_\pi^{fiscal} \in [0, 1)$, so the Taylor principle is not satisfied, and fiscal policy is active.

## B.5 Market clearing

The market clearing conditions for the domestic good in the home and foreign markets read:

$$Y_{Ht} = (1-\gamma)\, h\left(\frac{P_{Ht}^N}{\mathcal{P}_t}\right)(C_t + Z_t + X_t) \tag{B.26}$$

$$Y_{Et} = \gamma h\left(\frac{P_{Ht}^{N*}}{\mathcal{P}_t^*}\right)(C_t^* + Z_t^* + X_t^*)\,. \tag{B.27}$$

The model is closed with the resource constraint of the home economy:

$$C_t + Z_t + B_t - \frac{B_{t-1}R_{t-1}}{\Pi_t} = P_{Ht}Y_{Ht} + Q_t P_{Ht}^* Y_{Et} - X_t, \tag{B.28}$$

where the left-hand side includes domestic absorption (consumption plus investment) and the trade balance, which equals the net foreign asset position minus its return; the right-hand side is GDP, consisting of the value of home production net of intermediate inputs.

# C Additional figures and tables

Table C.1: Simple-Model comparison

| | **Wedge in the Backus-Smith condition** | **Incomplete financial markets** |
|---|---|---|
| *Floating regime (99Q1–19Q4)* | | |
| $corr(\Delta e_t, \Delta q_t)$ | 1.00 | 1.00 |
| $corr(\Delta e_t, \pi_t)$ | 0.09 | 0.11 |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | 18 | 14 |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 0.19 | 0.23 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 3.36 | 3.11 |
| *Pegged regime (94Q3–98Q4)* | | |
| $corr(\Delta e_t, \Delta q_t)$ | - | - |
| $corr(\Delta e_t, \pi_t)$ | - | - |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | - | - |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 1.46 | 1.10 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 1.45 | 1.09 |
| *Fiscal regime (80Q2–94Q2)* | | |
| $corr(\Delta e_t, \Delta q_t)$ | 0.62 | 0.62 |
| $corr(\Delta e_t, \pi_t)$ | 0.83 | 0.86 |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | 1.27 | 1.27 |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 4.78 | 4.78 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 3.36 | 3.11 |

Table C.2: Calibrated parameters in the quantitative model

| Parameter | Description | Value | Parameter | Description | Value |
|---|---|---|---|---|---|
| $\beta$ | Discount factor | 0.99 | $\bar{a}$ | Steady-state TFP | $-0.0310$ |
| $\sigma$ | Relative risk aversion | 2 | $\kappa$ | Investment adjustment cost | 6.3 |
| $\nu$ | Frisch elasticity | 1 | $\varsigma$ | Subsidy | 0.6667 |
| $n$ | Population share | 0.19 | $\Pi^*$ | Foreign inflation target | 1.005 |
| $\gamma$ | Home trade openness | 0.061 | $\rho_a, \rho_\psi, \rho_d$ | AR(1) parameter | 0.97 |
| $\gamma^*$ | Foreign trade openness | 0.0143 | $\sigma_a$ | SD of TFP shock | 1 |
| $\theta$ | Elasticity of substitution | 1.5 | $\sigma_\psi$ | SD of financial shock | 65 |
| $\vartheta$ | Share of capital in value added | 0.3 | $\sigma_d$ | SD of deficit shock | 1.483 |
| $\phi$ | Share of intermediate input | 0.5 | $\rho_{a,a^*}$ | TFP correlation | 0.33 |
| $\alpha$ | Strategic complementarity | 0.4 | $\Gamma$, $\iota$ | Fin. friction | 0.0043, 0.001 |
| $\delta$ | Depreciation rate | 0.02 | $\Gamma_{pegged}$, $\iota_{pegged}$ | Fin. friction (pegged) | 0.0003, 0.00007 |
| $\rho_m$ | Interest-rate smoothing | 0.95 | | | |

*Notes:* The standard deviation of the international financial shock remains unchanged across regimes. In the linearized UIP condition, this shock enters multiplied by $\Gamma$, which is close to zero when the exchange rate is pegged.

Table C.3: Parameters in the policy rules

| Regime | $\phi_\pi$ | $\phi_e$ | $\rho_m$ | $\phi_b$ | $\bar{\Pi}$ | $\bar{B}_G$ | $\bar{D}$ |
|---|---|---|---|---|---|---|---|
| *Floating* | 2.15 | 0 | 0.95 | 0.09 | 1.0075 | 0.80 | $-0.004$ |
| *Pegged* | 0.87 | 0.36 | 0.95 | 0.09 | 1.0075 | 0.80 | $-0.004$ |
| *Fiscal* | 0.8 | 0 | 0.95 | 0 | 1.0913 | 0.80 | 0.0169 |

Table C.4: Sensitivity analysis

| | **Baseline** | **Money supply** | **Deficit vol.** | **Same SS** | **Data** |
|---|---|---|---|---|---|
| *Fiscal regime (80Q2–94Q2)* | | | | | |
| $corr(\Delta e_t, \Delta q_t)$ | 0.57 | 0.51 | 0.83 | 0.55 | 0.55 |
| $corr(\Delta e_t, \pi_t)$ | 0.87 | 0.87 | 0.70 | 0.88 | 0.67 |
| $\sigma(\Delta e_t)/\sigma(\pi_t)$ | 1.20 | 1.15 | 1.77 | 1.19 | 1.16 |
| $\sigma(\pi_t)/\sigma(\Delta gdp_t)$ | 4.78 | 4.78 | 2.17 | 4.78 | 4.78 |
| $\sigma(\Delta q_t)/\sigma(\Delta gdp_t)$ | 2.64 | 2.55 | 2.64 | 2.65 | 4.26 |